

Proposed Problems

1. **212.** (Dan Radu)

In $\mathbb{R}[x]$ consider the polynomials

$$F_n(x) = x(x-1)\dots(x-n+1) = \sum_{k=1}^n S_n^k x^k,$$

where $S_n^1, \dots, S_n^n \in \mathbb{Z}$ are Stirling numbers of the first kind.

- Establish a relation between S_{n+1}^k , S_n^k and S_n^{k-1} , then give a general formula for S_n^k in terms of n and k .
- Show that any polynomial $P_n(x) = x^n + a_1 x^{n-1} + \dots + a_n$ can be represented as $P_n(x) = (-1)^n D_n(x)$ where $D_n(x)$ is the characteristic polynomial of a suitably chosen matrix $A_n \in M_n(\mathbb{R})$.
- Write the matrix A_n corresponding to $f_n(x) = (x-1)(x-2)\dots(x-n)$, diagonalise it, and derive a new method for computing the Stirling numbers $S_{n+1}^1, \dots, S_{n+1}^n$. Treat the case $n=4$.
- Consider $\varphi_n(x) = \frac{(-1)^n}{n!}(x-1)\dots(x-n)$ and $\pi_n(x) = 1 - \frac{x}{1!} + \dots + \frac{(-1)^n}{n!}x(x-1)\dots(x-n+1)$. Show that $\varphi_n = \pi_n$ and express their coefficients via Stirling numbers in two ways.
- Using the results of (d), prove

$$S_{n+1}^{k+1} = n! \sum_{p=k}^n \frac{(-1)^{n+p}}{p!} S_p^k, \quad S_n^{n-r} = (n-1)! \sum_{p=0}^r \frac{(-1)^p}{(n-p-1)!} S_{n-p-1}^{n-r-1}.$$

2. **213.** (Róbert Szász)

For every $n \in \mathbb{N}$, $n \geq 2$, prove

$$\prod_{i=0}^{n-1} \frac{(3i+1)}{n!} + \prod_{i=0}^{n-1} \frac{(3i+2)}{n!} + \sum_{k=1}^{n-1} \frac{\prod_{i=0}^{k-1} (3i+1) \prod_{j=0}^{n-k-1} (3j+2)}{k!(n-k)!} = 3^n.$$

3. **214.** (Nicolae Pavelescu)

For a fixed $s > 0$, without using derivatives, find

$$L(s) = \lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow 0} \frac{2 \left(1 - (\operatorname{trig}_1 x)^{\frac{1}{x+2}} (\operatorname{trig}_2 x)^{\frac{1}{(n-1)x+2}} \dots (\operatorname{trig}_n x)^{\frac{1}{x+2}} \right)}{x^2} \right),$$

where $\operatorname{trig} \in \{\cos, \cosh\}$.

4. **215.** (Vasile Cîrtoaje)

Let $x_1, \dots, x_n > 0$ with $x_1 x_2 \dots x_n = 1$. Prove that for $\frac{n-1}{n+1} \leq r < 1$,

$$\frac{x_1^r}{x_1 + x_2 + \dots + x_n} + \frac{x_2^r}{x_1 + x_2 + \dots + x_n} + \dots + \frac{x_n^r}{x_1 + x_2 + \dots + x_{n-1} + x_n} \geq 1.$$

5. **216.** (Răzvan Iaagăr)

Construct a sequence of continuous functions $f_n : [0, 1] \rightarrow [0, 1]$ such that

- $\int_0^1 f_n(x) dx \rightarrow 0$;
- $(f_n(x))_n$ does not converge for any $x \in (0, 1)$;
- each f_n takes every positive value at most twice.

6. **217.** (Marian Tetiva)

Let $x, y, z \in \mathbb{R}$ with $S = x + y + z > 0$. Prove

$$9((x+y+z)(xy+yz+zx) - 9xyz) \leq (\geq) 2((x+y+z)^3 - 27xyz),$$

where the inequality sign is $\leq$ if two of x, y, z are $\leq S/3$, and $\geq$ if two are $\geq S/3$.

7. **218.** (Gabriel Dospinescu)

Let $a, b, c \in \mathbb{Z}$ such that $b^2 - 4ac$ is not a perfect square. Prove that for every $n > 1$ there exist n consecutive natural numbers that cannot be represented in the form $(ax^2 + bxy + cy^2)z$ with $x, y, z \in \mathbb{Z}$, $z > 0$.

8. **219.** (Dan Radu)

Let G be a multiplicative group of matrices over a commutative field K .

- (a) Show that all matrices in G are square and have the same order.
- (b) Show that the identity of G need not coincide with the unit matrix, but if it does then G is a subgroup of $\text{GL}_n(K)$. Construct examples in $M_2(K)$.
- (c) Show that the matrices in G are either all nonsingular (then $G \leq \text{GL}_n(K)$) or all singular.
- (d) A matrix A is called a *group matrix* if there exists a group G with $A \in G$. Show that every idempotent matrix is a group matrix but no nilpotent matrix is. Decide which of the following are group matrices:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, D = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

- (e) Show that if A is a group matrix and B is similar to A , then B is also a group matrix.
- (f) Show that if A is a group matrix then $\text{rk } A = \text{rk } A^2$.

9. **220.** (Andrei Vernescu)

Find a real polynomial p that has a fixed point x_0 and commutes (under composition) with an injective function $f: \mathbb{R} \rightarrow \mathbb{R}$ that has no fixed points.

10. **221.** (Marian Tetiva)

Let $n \geq 1$. The set $\{n+1, n+2, \dots, 2n\}$ can be partitioned into four subsets with equal sum of elements if and only if n is a multiple of 8 or $n = 8k+5$ with $k \geq 1$.

11. **222.** (Vasile Cîrtoaje)

Let $a, b, c > 0$. Prove

$$\frac{a^4}{a^3+b^3} + \frac{b^4}{b^3+c^3} + \frac{c^4}{c^3+a^3} \geq \frac{a+b+c}{2}.$$

12. **223.** (Alexandru Lupuş and Andrei Vernescu)

- (a) Let $A(a)$ be the set of bounded real sequences (x_n) satisfying $x_n = nx_{n-1} - 1/a$ with $x_0 = 1 - 1/a$, $a > 0$. Determine a .
- (b) Let $B(a)$ be the set of bounded real sequences (x_n) satisfying $x_n = a - nx_{n-1}$ with $x_0 = a - 1$, $a > 0$. Determine a .

13. **224.** (Dan Radu)

Let E be a 4-dimensional vector space over $\mathbb{R}$ with basis $B = \{e_1, e_2, e_3, e_4\}$. The matrix of an endomorphism u in this basis is

$$B(u) = \begin{pmatrix} -2 & -3 & -1 & 1 \\ -2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -1 \\ -5 & -6 & -1 & 6 \end{pmatrix}.$$

- (a) Set $a_1 = e_1 - e_2$, $a_2 = e_2 + e_4$, $a_3 = e_3$, $F = \text{Sp}\{a_1, a_2, a_3\}$. Show that F is invariant under $\psi = 2\text{id}_E - u$ and that $v = \psi|_F$ is nilpotent of index 3.
- (b) Let $f_1 = 2e_1 - e_2 - e_3 + e_4$, $f_2 = -v(f_1)$, $f_3 = -v(f_2)$. Show that $\{f_1, f_2, f_3\}$ is a basis of F . Find α, β, γ such that $f_4 = e_1 + \alpha e_2 + \beta e_3 + \gamma e_4$ belongs to $\ker(3\text{id}_E - u)$.
- (c) Show that $C = \{f_1, f_2, f_3, f_4\}$ is a basis of E and write $C(u)$. Find the transition matrices between B and C , and compute $C^n(u)$ for $n \in \mathbb{N}$.
- (d) Consider $p(x) = x^4 - 9x^3 + 30x^2 - 44x + 24$. Show that $\theta_\lambda = \lambda\text{id}_E - u$ is an automorphism iff $p(\lambda) \neq 0$, and determine the rank of θ_λ as a function of λ .

14. **225.** (Mihály Bencze)

For all $n \geq 1$,

$$\left\lceil \frac{1}{n} \sum_{k=1}^n \frac{1}{\ln(1 + \frac{1}{k})} \right\rceil = \left\lfloor \frac{3n+5}{6} \right\rfloor,$$

where $\lceil \cdot \rceil$ denotes the integer part.

15. **226.** (José Luis Díaz-Barrero)

Compute

$$\lim_{n \rightarrow \infty} \left\{ n^2 \sum_{k=1}^n \frac{(-1)^{k+1}}{k+1} \binom{n}{k} \left[\left(\sum_{j=1}^k \frac{1}{j+1} \right)^2 + \sum_{j=1}^k \frac{1}{(j+1)^2} \right] \right\}.$$

16. **227.** (Ovidiu Pop)

Show that $\binom{n}{k} x^k (1-x)^{n-k} \leq \frac{1}{2}$ for every $x \in [0, 1]$ and every $k \in \{1, \dots, n-1\}$, $n \geq 2$.

17. **228.** (Dan Radu)

For $A \in M_3(\mathbb{R})$ define $A^* = J^t A J$ with $J = \text{diag}(1, 1, -1)$ and let $\Gamma = \{A \in M_3(\mathbb{R}) \mid A^* A = I\}$, $G = \{u \in L(\mathbb{R}^3) \mid M(u) \in \Gamma\}$.

- (a) Let G' be the set of $u \in G$ such that $u(e_1) = e_1$ and the component of $u(e_3)$ along e_3 is positive. Show that $M(u)$ has one of the two forms

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cosh \theta & \sinh \theta \\ 0 & \sinh \theta & \cosh \theta \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\cosh \theta & -\sinh \theta \\ 0 & \sinh \theta & \cosh \theta \end{pmatrix}, \quad \theta \in \mathbb{R}.$$

- (b) Show that $M(u)$ is diagonalisable and give its diagonal form and the transition matrices.

- (c) Prove that $A \in \Gamma$ implies $|\det A| = 1$ and $a_{33}^2 \geq 1$. Set $\Gamma' = \{A \in \Gamma \mid a_{33} \geq 1\}$ and show that Γ' is a subgroup of Γ .

18. **229.** (Marian Tetiva)

The set $\{1^2, 2^2, \dots, n^2\}$ can be partitioned into three subsets with equal sum of elements iff n is of the form $9k$, $9k-1$ or $9k-5$ with $k \geq 2$.

19. **230.** (Marian Tetiva, Marius Olteanu)

Let $x_1, \dots, x_n > 0$ and $r = \sqrt[n]{x_1 \cdots x_n} \geq (n-1)^{1-2/n}$. Prove

$$e^{-x_1} + e^{-x_2} + \dots + e^{-x_n} \geq n e^{-r}.$$

20. **231.** (Mihály Bencze)

If Γ denotes Euler's Gamma function, then for all $0 \leq x_1 < x_2 < \dots < x_n \leq 1$ and every positive integer $k \geq 1$,

$$\left(\frac{\Gamma(k+x_2)}{\Gamma(k+x_1)} \right)^{\frac{1}{x_2-x_1}} \left(\frac{\Gamma(k+x_3)}{\Gamma(k+x_2)} \right)^{\frac{1}{x_3-x_2}} \dots \left(\frac{\Gamma(k+x_n)}{\Gamma(k+x_{n-1})} \right)^{\frac{1}{x_n-x_{n-1}}} < (k+1)^{n-1}.$$

21. **232.** (Nicolae Pavelescu)

If $A \in M_n(\mathbb{C})$ and $A^m = (-1)^m I_n$ for some $m \geq 2$, then

$$n \leq \text{rk}(I_n + A) + \text{rk}(I_n + \varepsilon A) + \dots + \text{rk}(I_n + \varepsilon^{m-1} A) \leq (m-1)n,$$

where $\varepsilon = \cos \frac{2\pi}{m} + i \sin \frac{2\pi}{m}$.

22. **233.** (Constantin P. Niculescu and Andrei Vernescu)

Show that the primitive of $f(x) = \sin^p x$ on $(0, \pi/2)$ can be reduced to the primitive of a rational function in $\sin x$ iff $p \in \mathbb{N}$; moreover, then the primitive contains only a linear combination of natural powers of $\sin x$ and $\cos x$ and possibly a term in x .

23. **234.** (Dan Radu)

Let G be an additive abelian group and $f : G \rightarrow \mathbb{R}$ satisfy $f(x+y) + f(x-y) \leq 2f(x) + 2f(y)$ for all $x, y \in G$. Prove $f(nx) \leq n^2 f(x)$ for all $n \in \mathbb{N}$, $x \in G$. If every element of G has finite order, then f is positive.

24. **235.** (Marcel Tena)

Prove:

- (a) $\cos \pi/7$ has minimal polynomial $X^3 - \frac{1}{2}X^2 - \frac{1}{2}X + \frac{1}{8}$ over $\mathbb{Q}$.

- (b) $\cos 2\pi/7$ has minimal polynomial $X^3 + \frac{1}{2}X^2 - \frac{1}{2}X - \frac{1}{8}$.

- (c) $\cos 3\pi/7$ has the same minimal polynomial as $\cos \pi/7$.

- (d) $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$ and $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} = \frac{1}{8}$.

25. **236.** (Marian Tetiva)

Let A be a ring with unity such that for every $x, y \in A$ the binomial theorem holds for three consecutive positive integers n . Prove A is commutative. Does the conclusion remain true if A lacks unity?

26. **237.** (Marius Olteanu)

In an orthocentric tetrahedron $ABCD$ denote h_A, m_A the altitude and median from A to face BCD , and r, R the inradius and circumradius. Prove

$$\frac{1}{4r^2} \leq \frac{1}{h_A^2} + \frac{1}{h_B^2} + \frac{1}{h_C^2} + \frac{1}{h_D^2} \leq \frac{1}{324} \frac{R^4}{r^6}, \quad \frac{9}{4R^2} \leq \frac{1}{m_A^2} + \frac{1}{m_B^2} + \frac{1}{m_C^2} + \frac{1}{m_D^2} \leq \frac{1}{324} \frac{R^4}{r^6}.$$

27. **238.** (Dan Radu)

Let G be additive abelian, $f : G \rightarrow \mathbb{R}$ with $f(x+y) + f(x-y) \leq 2f(x) + 2f(y)$. Let $M = \{m > 1 \mid f(mx) = m^2 f(x) \forall x \in G\}$. If $M \neq \emptyset$ then $\inf M = 2$.

28. **239.** (Marian Tetiva)

Let q be a prime divisor of $F_n = 2^{2^n} + 1$ of the form $2^{n+2}k + 1$ with k odd and $q^2 \nmid F_n$. Then $2^{q-1} \not\equiv 1 \pmod{q^2}$ and the order of 2 modulo q^2 is divisible by q .

29. **240.** (Vasile Cîrtoaje)

If $a, b > 0$, $x = \sqrt{ab} \geq 1$, then for any $k \in \mathbb{N}$,

$$\frac{1}{1+a+\dots+a^k} + \frac{1}{1+b+\dots+b^k} \geq \frac{2}{1+x+\dots+x^k}.$$

30. **241.** (Gheorghe Szöllösy)

Find $\alpha \in \mathbb{R}$ such that $\lim_{n \rightarrow \infty} n^\alpha \int_0^1 \sin^{2n} \frac{\pi x}{2} dx \in (0, \infty)$ and compute the limit.

31. **242.** (Nicușor Minculete)

In any triangle:

$$r_a^n \sin \frac{A}{2} + r_b^n \sin \frac{B}{2} + r_c^n \sin \frac{C}{2} \geq \frac{3}{2} \left(\frac{p\sqrt{3}}{3} \right)^n, \quad 2(4R+r)^2(2R-r) \geq p^2(11R-4r).$$

32. **243.** (Dan Radu)

Find all integer solutions of $x^2 + y^2 + z^2 + w^2 = x^2 y^2 z^2$.

33. **244.** (Vasile Cîrtoaje)

For $a, b, c > 0$, $p > -2$,

$$\frac{2a^2 + (1+2p)bc}{b^2 + pbc + c^2} + \frac{2b^2 + (1+2p)ca}{c^2 + pca + a^2} + \frac{2c^2 + (1+2p)ab}{a^2 + pab + b^2} \geq \frac{3(3+2p)}{2+p}.$$

34. **245.** (Marian Tetiva)

For any $x, y, z > 0$,

$$x^2 y^2 z^2 \left(\frac{1}{x^6 + y^3 z^3} + \frac{1}{y^6 + z^3 x^3} + \frac{1}{z^6 + x^3 y^3} \right) \leq \frac{3}{2}.$$

35. **246.** (Dan Coma)

Let $a, b, c > 0$, $n \geq 4$. Then

$$\left(\frac{na}{b+c} + 1 \right) \left(\frac{nb}{c+a} + 1 \right) \left(\frac{nc}{a+b} + 1 \right) > (n+1)^2.$$

36. **247.** (Ovidiu Pop)

Let $a, b, c > 0$ such that after a permutation x, y, z we have $z \leq y \leq x \leq 8z$ and $8y \leq 27z$. Then

$$\max\{|\sqrt[3]{a} - \sqrt[3]{b}|^3, |\sqrt[3]{b} - \sqrt[3]{c}|^3, |\sqrt[3]{c} - \sqrt[3]{a}|^3\} \leq \frac{a+b+c}{3} - \sqrt[3]{abc}.$$

37. **248.** (Dan Radu)

Let $C^k(\mathbb{R})$ be the space of k -times differentiable real functions, $L(y) = y'' + qy' + ry$, $D(y) = y'$.

(a) Show $\ker L \subseteq C^\infty(\mathbb{R})$ and $D|_{\ker L}$ is an endomorphism.

- (b) For a basis $\{y_1, y_2\}$ of $\ker L$ define $\phi : \ker L \rightarrow \mathbb{R}^2$. Show D induces $f \in L(\mathbb{R}^2)$ making the diagram commute. Find condition for f to be an automorphism.
- (c) If A_f is the matrix of f in the canonical basis and p is the characteristic polynomial of $L(y) = 0$, compute $p(A_f)$.
- (d) Choose a convenient basis of $\ker L$ and write the general form of A_f . Special cases: (i) $q = 0, r = -1$; (ii) $q = r = 1$; (iii) $q = 2, r = 1$.
38. **249.** (Vasile Cîrtoaje)
If $a, b > 0$ and $a + b = a^n + b^n$ for some $n \geq 2$, then $a^{n+1} + b^{n+1} \leq 2$.
39. **250.** (Gabriel Dospinescu and Marian Tetiva)
Let $a, b, c \in \mathbb{Q}$, $a \neq 0$, $4ac - b^2$ a nonzero square in $\mathbb{Q}$. Construct an additive $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $af(f(x)) + bf(x) + cx = 0$ for all $x \in \mathbb{R}$.
40. **251.** (Mihai Miculiță and Marius Olteanu)
In a simplex with altitudes h_i , ex-sphere radii r_i and inradius r ,
- $$\sum \frac{r_i}{r_i - r} \geq (n-1) \sum \frac{r_i}{r_i + r} \geq \frac{n^2}{2}, \quad \sum \frac{h_i}{h_i - r} \geq \frac{n+1}{n-1} \sum \frac{h_i}{h_i + r} \geq (n-1) \sum \frac{h_i}{(n-2)h_i + r} \geq \frac{n^2}{n-1},$$
- $$\sum \frac{r_i}{(n-2)r_i - r} \geq \frac{n+1}{n-1} \sum \frac{r_i}{(n-2)r_i + r} \geq (n-1) \sum \frac{r_i}{(n-2)^2 r_i + r} \geq \frac{n^2}{(n-2)(n-1)}.$$
41. **252.** (Gheorghe Szöllösy)
Let $x_i > 0$, $S = \sum x_i$, $\sigma = \sum_{i_1 < \dots < i_k} x_{i_1} \cdots x_{i_k}$. Then
- $$\sum_{i_1 < \dots < i_k} \sqrt{x_{i_1} \cdots x_{i_k}} (S - x_{i_1} - \dots - x_{i_k}) \leq \sqrt{C_{n-1}^k} S \sigma.$$
42. **253.** (Dan Radu)
Let $f : (a, b) \rightarrow \mathbb{R}$ be continuous and strictly positive, and $\phi : \mathbb{R} \rightarrow (a, b)$ differentiable with $\phi'(x) = f(\phi(x))$. Prove ϕ is invertible.
43. **254.** (Marian Tetiva)
For $p \geq q \geq 1$,
- $$\sum_{j=1}^{p-q+1} \frac{2^{j-1}}{j} \binom{p-j}{q-1} = \sum_{k=0}^{\lfloor (p-q)/2 \rfloor} \frac{1}{2k+1} \binom{p}{q+2k}.$$
44. **255.** (Gheorghe Szöllösy)
- $$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{2k-1}{\sqrt{n^2 + 8k^2 - 8k + 4}}.$$
45. **256.** (Nicușor Minculete)
In any tetrahedron, for $\alpha \geq 1$, $2R^\alpha + 3^\alpha r^\alpha \geq 2^{-\alpha} \cdot 3^{1+5\alpha/6} V^{\alpha/3}$.
46. **257.** (Mihály Bencze)
- $$\sum_{k=1}^n \ln \frac{1 + \sqrt{k^2(k+1)^2 + 1}}{k(k+1)} \leq \frac{n}{n+1} \leq \sum_{k=1}^n \ln \frac{k^2 + k}{k^2 + k - 1}.$$
47. **258.** (Laurențiu Modan)
Let $I_n = \int_0^1 (x - x^2)^n dx$. Show (I_n) converges and find its limit.
48. **259.** (Dan Radu)
Let E be the space of C^∞ functions $\mathbb{R} \rightarrow \mathbb{C}$. For $f \in E$ define $\phi_t(x) = f(x+t)$ and $E(f) = \text{Sp}\{\phi_t \mid t \in \mathbb{R}\}$.
- (a) Find $\dim E(f)$ for $f_1(x) = e^x$, $f_2(x) = \sin x$, $f_3(x) = x$, $f_4(x) = xe^x$. Show that if $f(x) = p(x)e^{\alpha x}$ (polynomial times exponential) then $\dim E(f) < \infty$, and that for $f(x) = 1/(x^2 + 1)$ it is infinite.
- (b) Let $g_1, \dots, g_n \in E$ be linearly independent. Show there exist distinct $a_1, \dots, a_n$ with $\det(g_i(a_j)) \neq 0$.
- (c) If $\dim E(f) = n$ and $\{g_1, \dots, g_n\}$ is a basis of $E(f)$, prove $f(x+t) = h_1(t)g_1(x) + \dots + h_n(t)g_n(x)$ for some $h_i \in E(f)$.

- (d) Under the same hypothesis, show f satisfies a linear homogeneous ODE of order n with constant coefficients.
- (e) Using (d), determine all f with $\dim E(f) = 2$.

49. **260.** (Marian Tetiva)

Find all complex numbers of modulus 1 such that $|1 + z + \dots + z^n| \geq 1$ for every even n .

50. **261.** (Dumitru Bătinețu-Giurgiu)

Let (a_n) be positive with $\lim(a_{n+1} - a_n) = a \in \mathbb{R}$. Compute

$$\lim_{n \rightarrow \infty} \left(\ln \frac{e}{mn} + \sum_{k=1}^{mn} \frac{1}{k} - \gamma \right)^{a_n},$$

where $m \in \mathbb{N}^*$ and γ is Euler's constant.

51. **262.** (Marian Tetiva)

Let $X = \begin{pmatrix} t & u \\ v & w \end{pmatrix} \in M_2(\mathbb{C})$, $X^n = \begin{pmatrix} t_n & u_n \\ v_n & w_n \end{pmatrix}$, $a = t+w$, $b = tw-uv$. Prove the sequences $(t_n), (u_n), (v_n), (w_n)$ converge iff one of the following holds:

- $X = I_2$;
- $a = b + 1$ and $|b| < 1$;
- $|a|^2 + |a^2 - 4b| < 2(|b|^2 + 1) < 4$.

52. **263.** (Marius Olteanu)

Let G be the centroid of tetrahedron $ABCD$, r, R its inradius and circumradius, and $r_1, \dots, r_4$ the inradii of $GABC, GABD, GACD, GBCD$. Prove

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4} < \frac{R^2}{r^3}, \quad r_1 + r_2 + r_3 + r_4 > \frac{16}{R^2}, \quad r_1 + r_2 + r_3 + r_4 > \frac{8}{5}r.$$

53. **264.** (Dan Radu)

Let V be the space of polynomials of degree $\leq n-1$ over $\mathbb{C}$. For $a \in \mathbb{C}$ define $u_a(p(x)) = p(x+a)$.

- (a) Show $\{u_a \mid a \in \mathbb{C}\}$ is a subgroup of $\text{GL}(V)$ isomorphic to $\mathbb{C}$. Find a subgroup of $\text{GL}_n(\mathbb{C})$ isomorphic to $\mathbb{C}$.
- (b) For $a \neq 0$ define $p_0 = 1$, $p_1 = x/(1! \cdot a)$, $p_n = x(x-a) \cdots (x-(n-2)a)/((n-1)!a^{n-1})$. Show $\{p_0, \dots, p_{n-1}\}$ is a basis and write the matrix of $u_a - \text{id}$.
- (c) Find $\ker(u_a - \text{id})^k$ and $\text{im}(u_a - \text{id})^k$.
- (d) Show $(\alpha u_a + \beta u_b)^n = \alpha \iff \ker(\alpha u_a + \beta u_b) \neq \{0\} \iff \alpha + \beta = 0$.

54. **265.** (George Stoica)

Let (x_n) be positive, $x_n \rightarrow 0$, $\sum x_n = \infty$. Then there exists a subsequence (x_{n_k}) such that $\sum x_{n_k} = \infty$ and $\sum x_{n_k}^2 < \infty$.

55. **266.** (Marian Tetiva)

Let $a \geq b \geq c > 0$ and $b^2 \geq ac$. Show $\phi(x) = ((a^x + b^x + c^x)/3)^{1/x}$ is logarithmically concave on $(0, \infty)$.

56. **267.** (Gheorghe Szöllösy)

Prove $\tan \frac{n\pi}{2n+1} - 2 \tan \frac{(n-1)\pi}{2n+1} > \frac{\pi}{2}$ for all $n \in \mathbb{N}^*$.

57. **268.** (Daniel Văcărețu)

Let $ABCD A' B' C' D'$ be a cube of side a , $X \in AB$, $Y \in AD$, $Z \in AA'$ with $AX = \alpha$, $AY = \beta$, $AZ = \gamma$ ($\alpha, \beta, \gamma > a$) satisfying $\frac{1}{\alpha} + \frac{1}{\beta} > \frac{1}{a}$, etc. The plane (XYZ) meets edges $A'B', BB', BC', CD', DD', D'A'$ at M, N, P, Q, R, S . Prove equivalence of:

- (a) MQ, NR, PS are concurrent.
- (b) The spheres inside the two parts of the cube tangent to (XYZ) and the tri-rectangular trihedra at A and C' have equal radii.
- (c) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{2}{a}$.

58. **269.** (Radu Gologan)

Two consecutive natural numbers $n, n+1$ are called *adapted* if $[(n+1)\sqrt{3}] - [n\sqrt{3}] = 1$. What is the probability that two randomly chosen consecutive numbers are adapted?

59. **270.** (Marian Tetiva)

Let $a, b, c > 0$ with $abc = 1$ and $c = \min\{a, b, c\}$. Prove

$$a^3 + b^3 + c^3 - 3(ab + ac + bc) \geq c(a-b)^2 + c(a-c)(b-c).$$

60. **271.** (Róbert Szász)

For every $\theta \in \mathbb{R}$, $\sum_{n=0}^{\infty} \frac{\cos n\theta}{(n+1)^4} \geq \frac{7\pi^4}{720}$.

61. **272.** (Dorin Mărghidanu)

Let $0 \leq x_1 \leq x_2 \leq \dots \leq x_n$ and $\sum x_i = a$. Determine the set $\{\max x_i \mid i = 1, \dots, n\}$ and specify when these maxima are attained.

62. **273.** (Dumitru Bătinețu-Giurgiu)

Let $e(x) = (1 + 1/x)^x$ for $x \in \mathbb{R}$, $a \in \mathbb{R}_+$. Compute

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow \infty} \left[\left(\frac{x}{n} \right)^2 \left(\sum_{k=1}^n e(x+ka) - ne(x) \right) \right] \right).$$

63. **274.** (Doru Ștefănescu)

Let $R, \rho > 0$ and $k \in \mathbb{N}$. Define $F_n(X) = X^n - R^k X^{n-k} - \sum_{\substack{1 \leq j \leq n \\ j \neq k}} \rho^j X^{n-j}$ for $n > k$.

(a) Show each F_n has a unique positive root ξ_n .

(b) Prove (ξ_n) is monotonic and convergent.

275. Let $A(z) = z^n + \sum_{k=0}^{n-1} a_k z^k$ be a polynomial with complex coefficients of degree $n \geq 2$ that has n distinct zeros $z_1, z_2, \dots, z_n$ and let α be a nonzero complex number such that no ratio of two zeros of $A(z)$ is equal to α . Prove that

$$\sum_{k=1}^n \left(\frac{1}{A(\alpha z_k) A'(z_k)} + \frac{1}{\alpha A(z_k/\alpha) A'(z_k)} \right) = 0.$$

276. Let d be a positive squarefree integer and (x_n, y_n) be the sequence of solutions of Pell's equation $x^2 - dy^2 = 1$. Show that for any positive integer N , there are infinitely many pairs of distinct positive integers m, n such that $\gcd(x_m, x_n) > N$.

277. If $u \in C^1([0, 1])$ and $u(0) = 0$, $u'(0) = 1$, then

$$\int_0^1 e^{u(x)} dx + \int_0^1 (u'(x))^2 dx \geq 4.$$

278. Prove that there does not exist a rational function R with real coefficients such that

$$R(n) = \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2}$$

for infinitely many natural numbers n .

279. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous and strictly increasing function, twice differentiable in 0 with $f(0) = f'(0) = 0$, $f''(0) > 0$, $\lim_{x \rightarrow \infty} f(x) = \infty$ and $a > 0$. Show that for all $n \in \mathbb{N}$ the equation

$$\sum_{k=1}^n f\left(\frac{kx}{n^2}\right) = a$$

has a unique solution x_n in $(0, \infty)$, and

$$\lim_{n \rightarrow \infty} \frac{x_n}{\sqrt{n}} = \sqrt{\frac{6a}{f''(0)}}.$$

280. Let $n \geq 2$ be a natural number. Determine the greatest positive number C such that the inequality

$$2^{n-1}(x^n + y^n) - (x + y)^n \geq C[(x + 3y)^n + (3x + y)^n - 2^{n+1}(x + y)^n]$$

holds for all $x, y \geq 0$.

281. If $u \in C^1([0, 1])$ and $u(0) = 0$, $u'(0) = 1$, then show that

$$\int_0^1 e^{u(x)} dx + \int_0^1 (u'(x))^2 dx \geq 24.$$

282. Denote $E_n(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}$ for all $n \in \mathbb{N}$, $x \in \mathbb{R}$. Prove that

$$(a) \quad \sum_{n=0}^{\infty} (e^x - E_n(x)) = xe^x; \quad (b) \quad \sum_{n=0}^{\infty} n(e^x - E_n(x)) = \frac{x^2}{2}e^x.$$

283. Let $f : \mathbb{R}_+^* \rightarrow \mathbb{R}_+^*$ be a function such that

$$\lim_{x \rightarrow \infty} \frac{(f(x))^{1/x}}{x} = a \in \mathbb{R}_+^*$$

and $s, t \in \mathbb{R}$ with $s + t = 1$. Then

$$\lim_{x \rightarrow \infty} ((x+1)^s (f(x+1))^{t/(x+1)} - x^s (f(x))^{t/x}) = b \in \mathbb{R}$$

exists if and only if

$$\lim_{x \rightarrow \infty} \frac{f(x+1)}{xf(x)} = c \in \mathbb{R}_+^*$$

and $a^t(s + t \ln(c/a)) = b$.

284. In the orthocentric tetrahedron $[ABCD]$ denote by r_A, r_B, r_C, r_D the inradii of faces BCD, ACD, ABD, ABC , and by r, R the inradius and circumradius of the tetrahedron. Show that the following refinement of Euler–Durrande inequality ($R \geq 3r$) holds:

$$R^2 \geq r^2 + r_A^2 + r_B^2 + r_C^2 + r_D^2 \geq 9r^2.$$

285. Prove that for all sequences (x_k) of positive numbers,

$$\sum_{k=1}^{\infty} \frac{x_k}{1 + (x_1 + x_2 + \cdots + x_{k-1})^2} > \frac{\pi}{2}.$$

In addition, if $x_1 + x_2 + \cdots + x_n = 1$, then

$$\sum_{k=1}^{\infty} \frac{x_k}{1 + (x_1 + x_2 + \cdots + x_{k-1})^2} > \frac{\pi}{4},$$

and these constants are best possible.

286. Let $f : (0, \infty) \rightarrow (0, \infty)$ be strictly decreasing and differentiable, and let F be a primitive of f . Suppose

$$(i) \quad \lim_{n \rightarrow \infty} \frac{f(n+1)}{f(n)} = 1, \quad (ii) \quad \lim_{n \rightarrow \infty} F(n) = 0, \quad (iii) \quad \frac{f'}{f} \text{ is strictly increasing.}$$

Then (a) the sequence $x_n = f(1) + f(2) + \cdots + f(n)$ is convergent; (b) the sequence $\frac{x - x_n}{F(n)}$ (where $x = \lim x_n$) is convergent and strictly monotone.

287. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous such that

$$\int_a^t f(x) dx \int_t^b f(x) dx \neq 0 \quad \forall t \in (a, b).$$

Show that there exists $c \in (a, b)$ such that

$$(c - a)f(c) \left(\int_c^b f(x) dx - \int_a^c f(x) dx \right) = \frac{1}{2} \int_a^c f(x) dx \int_c^b f(x) dx.$$

288. Let $p, q \geq 2$ be integers with $\gcd(p, q) = 1$. Show that $\log_p q$ is transcendental.
289. Find all plane curves such that the radius vector of a point makes a constant angle α ($0 < \alpha < \pi/2$) with the tangent at that point.
290. Study the convergence of the sequence defined by

$$x_1 = a, \quad 1 + x_n = (1 + x_{n-1})^{x_{n-1}}, \quad n \geq 1.$$

291. For $n, k \in \mathbb{N}$, $n \geq 1$, determine the dimension of the linear space of polynomials in n variables over a field K , of degree at most k , as a subspace of $K[x_1, x_2, \dots, x_n]$.
292. Let $n \geq 1$ be an integer, $I \subseteq \mathbb{R}$ an interval and $f : I \rightarrow \mathbb{R}$ a function.

- (a) Assume f is $(n-1)$ -times differentiable on I and $f^{(n-1)}$ is increasing on I . Prove that for all $a \leq b$ in I ,

$$f(b) - \binom{n}{1} f\left(\frac{(n-1)b+a}{n}\right) + \binom{n}{2} f\left(\frac{(n-2)b+2a}{n}\right) - \dots + (-1)^n f(a) \geq 0.$$

- (b) Assume f is n -times differentiable and $f^{(n)}$ is continuous on I . If for all $a \leq b$ in I we have

$$\sum_{k=0}^n (-1)^k \binom{n}{k} f\left(\frac{(n-k)b+ka}{n}\right) \geq 0,$$

then $f^{(n)}$ is nonnegative.

293. Prove that for any continuous function $f : [-1, 1] \rightarrow \mathbb{R}$,

$$\int_{-1}^1 f^2(x) dx \geq \frac{5}{2} \left(\int_{-1}^1 x^2 f(x) dx \right)^2 + \frac{3}{2} \left(\int_{-1}^1 x f(x) dx \right)^2.$$

294. Let $S(O, R)$ be the circumscribed sphere of tetrahedron $[ABCD]$ and r the inscribed radius. If x, y, z, t are the normal coordinates of O , then

$$x + y + z + t \leq 4\sqrt{R^2 - 8r^2}.$$

295. Determine all nonconstant polynomials $P \in \mathbb{Z}[X]$ such that $P(p)$ is square-free for all prime numbers p .

296. Let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n > 0$ and $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ be real numbers. Show that

$$\left(\sum_{1 \leq i, j \leq n} x_i x_j \min(a_i, a_j) \right) \left(\sum_{1 \leq i, j \leq n} y_i y_j \min(b_i, b_j) \right) \geq \left(\sum_{1 \leq i, j \leq n} x_i y_j \min(a_i, b_j) \right)^2.$$

297. Let S^2 be the two-dimensional sphere and $\alpha > 0$. Show that for any positive integer n and arbitrary points A_i, B_i, C_i on S^2 ($i = 1, \dots, n$), there exists $P_n \in S^2$ such that

$$\sum_{i=1}^n P_n A_i^\alpha = \sum_{i=1}^n P_n B_i^\alpha = \sum_{i=1}^n P_n C_i^\alpha$$

if and only if $\alpha = 2$.

298. Let t be an odd number. Find all monic polynomials $P \in \mathbb{Z}[X]$ such that for every integer n there exists an integer m with $P(m) + P(n) = t$.

299. Find all functions $f : \{1, 2, \dots\} \rightarrow \{1, 2, \dots\}$ satisfying:

$$(i) \ a - b \mid f(a) - f(b) \ \forall a, b; \quad (ii) \ \gcd(a, b) = 1 \Rightarrow \gcd(f(a), f(b)) = 1.$$

300. Consider the sequence $(a_n)_{n \geq 1}$ defined by $a_1 = 2$ and

$$a_{n+1} = 2a_n + \sqrt{3(a_n^2 - 1)}.$$

Show that all a_n are positive integers and that any odd prime p divides $a_p - 2$.

301. Determine the greatest prime number $p = p(n)$ such that there exists a matrix $X \in \text{SL}(n, \mathbb{Z})$ with $X^p = I_n$ and $X \neq I_n$.
302. Let $n \geq 2$ and denote by $\mathcal{D} \subset M_n(\mathbb{C})$ and $\mathcal{C} \subset M_n(\mathbb{C})$ the sets of diagonal and circulant matrices, respectively. Consider

$$\mathcal{V} = [D, C] = \text{span}\{dc - cd \mid d \in \mathcal{D}, c \in \mathcal{C}\}.$$

Prove that $\mathcal{C}, \mathcal{D}, \mathcal{V}$ span $M_n(\mathbb{C})$ if and only if n is prime.

303. Prove that

$$\sum_{n=0}^{\infty} \frac{1}{\Gamma(n+3/2)} = \frac{2e}{\sqrt{\pi}} \int_0^1 e^{-x^2} dx, \quad \sum_{n=0}^{\infty} \frac{\Gamma(-n+1/2) \Gamma(n+1/2)}{\Gamma(n+3/2)} = \frac{\sqrt{\pi}}{2e} \int_0^1 e^{x^2} dx.$$

304. Let $f, g : [0, 1] \rightarrow \mathbb{R}$ be such that f is continuous, g is increasing and differentiable with $g(0) \geq 0$. Prove that if

$$\int_t^1 f(x) dx \geq \int_t^1 g(x) dx \quad \forall t \in [0, 1],$$

then $\int_0^1 f^2(x) dx \geq \int_0^1 g^2(x) dx$.

305. Let K be a field and $f \in K[X]$ with $\deg f = n \geq 1$ having distinct roots $x_1, \dots, x_n$. For $p \in \{1, \dots, n\}$ let $S_1, \dots, S_p$ be the elementary symmetric polynomials in $x_1, \dots, x_p$. Show that

$$[K(S_1, S_2, \dots, S_p) : K] \leq \binom{n}{p}.$$

306. Let $x_1, \dots, x_n, y_1, \dots, y_n > 0$ with $\sum_{i=1}^n x_i = 1$. Denote by $\mathcal{A}$ the set of pairs (i, j) such that

$$y_1 + \dots + y_j \leq x_1 + \dots + x_i.$$

Prove that

$$\sum_{(i,j) \in \mathcal{A}} \frac{x_{i+1}y_j}{1 + (x_1 + \dots + x_{i+1})(y_1 + \dots + y_j)} \leq \frac{\pi^2}{24},$$

and the constant $\pi^2/24$ is best possible.

309. Consider a prime p and a rational number a with $\sqrt{a} \notin \mathbb{Q}$. Define the sequence of polynomials by $f_1 = X^p - a$ and $f_{n+1} = f_n^p - a$ for all $n \geq 1$. Show that all terms of the sequence f_n are irreducible.

310. For $n \geq 4$, $1 \leq \delta < \Delta \leq n-1$, $n, \delta, \Delta \in \mathbb{N}$, consider the function

$$f(x_\delta, x_{\delta+1}, \dots, x_\Delta) = \sum_{\delta \leq i < j \leq \Delta} \left(\frac{1}{\sqrt{i}} - \frac{1}{\sqrt{j}} \right)^2 x_i x_j$$

and the domain

$$D = \{(x_\delta, \dots, x_\Delta) : x_i \in \mathbb{N}, \sum_{i=\delta}^{\Delta} x_i = n\}.$$

Show that for $(x_\delta, \dots, x_\Delta) \in D$,

$$f(x_\delta, \dots, x_\Delta) \leq \left(\frac{1}{\sqrt{\delta}} - \frac{1}{\sqrt{\Delta}} \right)^2 \alpha(n),$$

where $\alpha(n) = n^2/4$ for even n and $(n^2 - 1)/4$ for odd n . When does equality hold?

311. Show that for any matrix $A \in M_2(\mathbb{R})$ there exist $X, Y \in M_2(\mathbb{R})$ with $XY = YX$ such that

$$A = X^{2n+1} + Y^{2n+1} \quad \text{for all } n \geq 1.$$

312. Let p_n be the n -th prime number. Show that the sequence $(x_n)_{n \geq 1}$ defined by

$$x_n = \left\{ \frac{1}{p_1} + \frac{1}{p_2} + \dots + \frac{1}{p_n} \right\} - \{\log \log n\}$$

is divergent. (Here $\{x\}$ denotes the fractional part of x .)

313. Does there exist a set M of points in the Euclidean plane such that the distance between any two of them is larger than 1 and such that there is a point in M between any two distinct parallel lines in the plane? Justify your answer.

314. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a C^2 real-valued function which is convex on $[0, 1]$. Prove

$$\int_0^1 f(x) dx \leq \frac{1}{4} \left(f(0) + f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right) + f(1) \right).$$

315. Let $f : [0, 2\pi] \rightarrow \mathbb{R}$ be such that

$$\int_0^{2\pi} f(x) \cos(kx) dx = 1 \quad \text{for all } k = 1, 2, \dots, n \ (n \geq 2).$$

Find the minimum of $\int_0^{2\pi} f^2(x) dx$ over all such functions f .

316. Let $f : [0, 1] \rightarrow [0, \infty)$ and $g : [0, 1] \rightarrow [0, 1]$ be integrable functions. Prove that for any positive integers p, q, s, t with $p \neq q$,

$$\int_0^1 f^p(x) g^s(x) dx \int_0^1 f^q(x) g^t(x) dx \leq \int_0^1 f^{p+q}(x) g^{\frac{sp-qt}{p-q}}(x) dx.$$

317. For integer $n \geq 2$, determine the dimension of

$$V = \text{span} \left\{ \frac{P(x)}{1 - x^{\deg P + 1}} : P(x) \in \mathbb{R}[x], 0 \leq \deg P < n - 1, \deg P + 1 \mid n \right\}$$

as a subspace of the $\mathbb{R}$ -linear space $\mathbb{R}(x)$.

318. Let f be a polynomial with integer coefficients, $\deg f \geq 1$, and let k be a positive integer. Show that there are infinitely many positive integers n such that $f(n)$ can be written as

$$f(n) = d_1 d_2 \dots d_k d_{k+1},$$

where $1 \leq d_1 < d_2 < \dots < d_k < n$.

319. Let μ be the Möbius function, $q \geq 1$ and $n \geq 2$. Define

$$N_j = \frac{1}{j} \sum_{d \mid j} \mu\left(\frac{n}{d}\right) q^d \quad (j \geq 1).$$

Show that

$$\sum_{x_1 + 2x_2 + \dots + nx_n = n} \binom{N_1}{x_1} \binom{N_2}{x_2} \dots \binom{N_n}{x_n} = q^n - q^{n-1}.$$

320. For $n > 1$, does there exist a quadratic polynomial $f \in \mathbb{Q}[X]$ such that $f^{2^n} + 1$ is reducible over $\mathbb{Q}$?

321. Find the probability that by choosing a positive integer n , the numbers $n\sqrt{2}$ and $n\sqrt{3}$ have even integral parts both.

322. Let K be an algebraically closed field and let

$$P = aX_1^{i_1} \dots X_n^{i_n} + bX_1^{j_1} \dots X_n^{j_n} + cX_1^{k_1} \dots X_n^{k_n} \in K[X_1, \dots, X_n]$$

with $abc \neq 0$. Assume that $X_t \nmid P$ for all t and that the exponent vectors $(i_1, \dots, i_n)$, $(j_1, \dots, j_n)$, $(k_1, \dots, k_n)$ are non-collinear. Prove that P is reducible if and only if $\text{char } K = p$ and $p \mid i_t, j_t, k_t$ for all t for some prime p .

337. (Originally 323) If m, n are given positive integers and A, B, C are three matrices of size $m \times n$ with real entries, then

$$\sum_{\text{cyc}} (\det(AB^T))^2 \det(CC^T) \leq \prod \det(AA^T) + 2 \prod_{\text{cyc}} |\det(AB^T)|.$$

338. (Originally 324) Let p be a prime number and a, b, c, d positive integers such that $a \geq c$ and $b, d \in \{0, 1, \dots, p-1\}$. Show that

$$\binom{ap+b}{cp+d} \equiv (a-c) \binom{a}{c} \binom{p+b}{d} + c \binom{a}{c} \binom{p+b}{p+d} - (a-1) \binom{a}{c} \binom{b}{d} \pmod{p^2}.$$

339. (Originally 325) Let p be a prime number. Show that

$$\sum_{k=1}^p \sqrt[k]{k + \sqrt[k]{k}}$$

cannot be rational.

340. (Originally 326) Let $A, B \in M_n(\mathbb{R})$ be diagonalisable in $M_n(\mathbb{R})$ such that $\exp(A) = \exp(B)$. Show that $A = B$.

341. (Originally 327) Let N be the $n \times n$ matrix with all entries equal to $\frac{1}{n}$ and $A \in M_n(\mathbb{R})$ such that $A^k = N$ for some positive integer k . Show that

$$\sum_{1 \leq i, j \leq n} a_{ij}^2 \geq 1.$$

342. (Originally 328) Given $a > 0$, let f be a real-valued continuous function on $[-a, a]$ and twice differentiable on $(-a, a)$. Show that for all $|x| < a$, there exists $|\xi| < x$ such that

$$f(x) + f(-x) - 2f(0) = x^2 f''(\xi).$$

343. (Originally 329) Let ABC be a triangle and let P be a point in its interior with pedal triangle DEF . Suppose that the lines DE and DF are perpendicular. Prove that the isogonal conjugate of P is the orthocenter of triangle AEF .

344. (Originally 330) It is well-known that for $p > 2$ prime, the number

$$N = \frac{2^{p-1} - 1}{p}$$

is integer. When is N a natural power of an integer?

345. (Originally 331) Let $f \in \mathbb{Z}[X]$ be a monic polynomial of degree $n + 2$, with $f(0) \neq 0$, $n \in \mathbb{N}$, $n \geq 1$. Show that there are only finitely many positive integers a such that $f(X) + aX^n$ is reducible over $\mathbb{Z}[X]$.

346. (Originally 332) The cells of a rectangular $2011 \times n$ array are colored using two colors, so that for any two columns the number of pairs of cells situated on the same row and bearing the same color is less than the number of pairs of cells bearing different colors.

(i) Prove that $n \leq 2012$. (ii) For a square array ($n = 2011$) each color appears at most $1006 \cdot 2011$ (hence at least 1

347. (Originally 333) Prove that for any $m, n \geq 3$ there is an $m \times n$ matrix of rank 2 with distinct prime entries.

348. (Originally 334) Define

$$\mathcal{F} = \{f : [0, 1] \rightarrow [0, 1] \mid \exists A, B \subset [0, 1], A \cap B = \emptyset, A \cup B = [0, 1], f(A) \subset B, f(B) \subset A\}.$$

Prove that $\mathcal{F}$ contains functions with the Darboux property.

349. (Originally 335) Let $f : [0, 1] \rightarrow \mathbb{R}$ be an integrable function with $\int_0^1 f(x) dx = 0$. Prove that

$$\int_0^1 f^2(x) dx \geq 12 \left(\int_0^1 x f(x) dx \right)^2.$$

350. (Originally 336) Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$, denote by f^n its n -th iterate. It is also given that $|f(x) - f(y)| \leq |x - y|$ for all $x, y \in \mathbb{R}$ and that $f^N(0) = 0$ for some $N \in \mathbb{N}^*$.

(i) If N is odd, prove that $|f(x)| \leq |x| \forall x \in \mathbb{R}$.

(ii) If N is even, prove that $|f(f(x))| \leq |x| \forall x \in \mathbb{R}$, but not necessarily $|f(x)| \leq |x|$.

351. Let $(a_n)_{n \geq 1}$ be a sequence of positive integers and let $\alpha > \frac{1}{2}$ such that $\sum_{n \geq 1} a_n^{-\alpha} = \infty$. Prove that for any $k \geq 2$ there is an integer that can be represented in at least k ways as a sum of two elements of the sequence.

352. Let K be a field and let m, n, k be positive integers. Find necessary and sufficient conditions on integers a, b, c such that there exist matrices $A \in M_{m,n}(K)$ and $B \in M_{n,k}(K)$ with $\text{rank } A = a$, $\text{rank } B = b$ and $\text{rank } AB = c$.

353. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be continuous and differentiable at 0. Denote

$$I(h) = \int_{-h}^h f(x) dx, \quad h \in [0, 1].$$

Show that

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n \phi(k) k |I(1/k)| = \frac{6}{\pi^2} |f(0)|,$$

where ϕ is Euler's totient function.

354. For $x > 1$, define $f(x) = \int_1^\infty e^{itx} dt$. Prove that there exists $L \in \mathbb{C}^*$ such that $\lim_{x \rightarrow \infty} xf(x) = L$.

355. Let p be an odd prime and $\alpha \in [0, \pi/2]$ such that $\cos \alpha = 1/p$. Prove that for any integer $n > 1$ there is no integer m such that $\cos(n\alpha) = 1/m$.

356. Let $\{b_n\}_{n \geq 0}$ be a sequence of positive real numbers. The following statements are equivalent:

$$\begin{aligned} \text{(i)} \quad & \sum_{n=0}^{\infty} \frac{|b_{n+1} - b_n|}{b_n^r} < \infty \quad \forall r \in \mathbb{R}; \quad \text{(ii)} \quad \sum_{n=0}^{\infty} \frac{|b_{n+1} - b_n|}{b_n} < \infty; \quad \text{(iii)} \quad \sum_{n=0}^{\infty} |b_{n+1} - b_n| < \infty \text{ and } \lim b_n > 0; \\ \text{(iv)} \quad & \sum_{n=0}^{\infty} \frac{|b_{n+1} - b_n|}{b_{n+1}} < \infty; \quad \text{(v)} \quad \sum_{n=0}^{\infty} \frac{|b_{n+1} - b_n|}{b_{n+1}^r} < \infty \quad \forall r \in \mathbb{R}. \end{aligned}$$

357. Describe the functions $\phi : \mathbb{R} \rightarrow \mathbb{R}$ with $\phi(0) = 0$ such that the set $\{\phi + y \mid y \in \mathbb{R}\}$ is a semigroup with respect to composition. Prove that such a semigroup is a monoid if and only if ϕ is the identity map.

358. Prove that for any coloring of the lattice points of the plane with a finite number of colors and for any triangle ABC having angles with rational tangents, there is a triangle with lattice vertices of the same color which is similar to ABC .

359. Determine how many permutations of the 81 squares of the Sudoku grid have the property that for any solution of the Sudoku game, applying the permutation to the 81 squares we obtain another solution of the Sudoku game.

360. Let $M_n(\mathbb{C})$ be the ring of square matrices of size n and $A \in M_n(\mathbb{C})$. Show that if for all positive integers k we have $\det((\text{adj } A)^k + I_n) = 1$, then $(\text{adj } A)^2 = O_n$.

361. 88% of the surface of a sphere is colored red. Prove that there is a cube inscribed in the sphere with all vertices red.

362. Given a function $f : X \rightarrow X$, denote

$$f_0(X) = X, \quad f_n(X) = f(f_{n-1}(X)), \quad f_\omega(X) = \bigcap_{n \geq 0} f_n(X).$$

(i) $f(f_\omega(X)) \subseteq f_\omega(X)$. (ii) For $X = \mathbb{R}$ and f continuous, $f_\omega(\mathbb{R})$ is $\mathbb{R}$, a half-line, a bounded segment, a singleton, or $\emptyset$. Moreover, assume $f(f_\omega(\mathbb{R})) = f_\omega(\mathbb{R})$.

(iii) If $f_\omega(\mathbb{R})$ is bounded, then it is a closed interval (possibly degenerate). Give examples.

(iv) Give an example for $f_\omega(\mathbb{R})$ being an open half-line.

363. For a given sequence $(x_n)_{n \geq 1}$ of real numbers and a fixed positive integer n_0 , consider the conditions:

$$(C_1) \quad n^2(x_{n+1} - x_n) - (2n+1)x_n \text{ has the same sign for all } n \geq n_0;$$

$$(C_2) \quad x_{n+m} \leq x_n + x_m \text{ for all } n, m \neq n_0;$$

$$(C_3) \quad \sum_{n=1}^{\infty} n^{-2} x_n < \infty; \quad (C_4) \quad \lim_{n \rightarrow \infty} \frac{x_n}{n} = 0.$$

Prove that:

(a) none of $(C_1), (C_2), (C_3)$ implies (C_4) ; (b) (C_4) follows from (C_3) and either (C_1) or (C_2) ;

(c) the converse of (b) is false.

364. Let $(x_n)_{n \geq 1}$ be a sequence of real numbers such that

$$\limsup_{n \rightarrow \infty} (1 - x_n) \log n < \infty.$$

Show that if the series $\sum_{n \geq 1} a_n$ (with $a_n > 0$) converges, then $\sum_{n \geq 1} a_n^{x_n}$ also converges.

365. Let K be a field and let $f, g \in K[X]$, $f, g \notin K$, such that $g^n - 1 \mid f^n - 1$ for all $n \geq 1$. Then f is a power of g .

366. Let K be an algebraically closed field of characteristic $p > 0$. For $i \geq 0$ define $Q_i \in \mathbb{Q}[X]$ by $Q_0 = X$ and $Q_{i+1} = \frac{Q_i^p - Q_i}{p}$. If $k \geq 0$ writes in base p as $k = c_0 + c_1 p + \cdots + c_s p^s$ with $0 \leq c_i \leq p - 1$, define $P_k = Q_0^{c_0} Q_1^{c_1} \cdots Q_s^{c_s}$. Prove that if

$$f = X^k + a_{k-1} X^{k-1} + \cdots + a_0 \in K[X] \quad (a_0 \neq 0)$$

has roots $\alpha_1, \dots, \alpha_s$ with multiplicities $k_1, \dots, k_s$, then

$$V_f := \{(x_n)_{n \geq 0} \mid x_n \in K, x_{n+k} + a_{k-1} x_{n+k-1} + \cdots + a_0 x_n = 0 \ \forall n \geq 0\}$$

is a vector space with basis $\{(P_j(n) \alpha_i^n)_{n \geq 0} \mid 1 \leq i \leq s, 0 \leq j \leq k_i - 1\}$.

367. Give examples of functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that f has period $\sqrt{2}$, g has period $\sqrt{3}$, and $f + g$ has period $\sqrt{5}$.

368. Find all matrices $X_1, \dots, X_9 \in M_2(\mathbb{Z})$ with $\det X_k = 1$ for all k and

$$X_1^4 + \cdots + X_9^4 = X_1^2 + \cdots + X_9^2 + 18I_2.$$

369. A stick is broken at random at two points (each point is uniformly distributed relative to the whole stick) and the parts' lengths are denoted by r, s, t . Show that the probability of the existence of a triangle encompassing three circles of radii r, s, t , each side tangent to two of the circles, and the circles mutually externally tangent, is $\frac{5}{27}$.

370. Calculate the improper integral $\int_0^\infty \cos^2 x \cos x^2 dx$.

371. Let $[ABCD]$ be a Crelle tetrahedron and let M, N, P, Q, R, S be the contact points of the sphere tangent to its edges. Prove that $V_{[MNPQRS]} \leq \frac{1}{2} V_{[ABCD]}$ (where V_X denotes volume).

372. Prove that $\lim_{n \rightarrow \infty} e^{-n} \left(1 + n + \frac{n^2}{2!} + \cdots + \frac{n^n}{n!} \right) = \frac{1}{2}$.

373. Let $n \geq 1$ and let $\Phi_n(X, q) = \prod_{k=1}^n (X - q^{2k-1}) = a_0 + a_1 X + \cdots + a_n X^n$ with $a_i \in \mathbb{R}[q]$. Prove that

$$\frac{\sum_{i=0}^{n-1} a_i a_{i+1}}{\sum_{i=0}^n a_i^2} = \frac{-q(1 - q^{2n-1})}{1 - q^{2n+1}}.$$

374. Let $A_1, A_2, \dots, A_n$ be points in the three-dimensional Euclidean space. Prove that on the unit sphere S^2 there is a point P such that $PA_1 \cdot PA_2 \cdots PA_n \geq 1$.

375. Let $n \geq 3$ be an integer. Find effectively the isomorphism class of the Galois group $\text{Gal}(\mathbb{Q}(\cos \frac{2\pi}{n})/\mathbb{Q})$.

376. (a) Show that the probability of a point $P(x, y, z)$, chosen at random with uniform distribution in $[0, 1]^3$, to be at distance at most $\sqrt{2}$ from the origin is $\frac{(15-8\sqrt{2})\pi}{12}$.

$$(b) \text{ Prove that } \int_0^{\pi/4} \frac{\cos^{3/2} 2\theta}{\cos^3 \theta} d\theta = \frac{(4\sqrt{2} - 5)\pi}{4}.$$

377. Let $p > 2$ be a prime and n a positive integer. Prove that

$$p^{\lfloor \frac{n-1}{p-1} \rfloor} \left| \sum_{k=0}^{\lfloor n/p \rfloor} (-1)^k \binom{n}{pk} \right|.$$

378. Let $(x_n)_{n \geq 1}$ be a sequence with $0 < x_n < 1$. Then the following are equivalent:

- (i) For any convergent series $\sum a_n$ ($a_n > 0$), the series $\sum a_n^{x_n}$ converges.
- (ii) The series $\sum_{n \geq 1} M^{-1/(1-x_n)}$ converges for some $M > 1$ large enough.

379. For any prime p we denote by $|\cdot|_p : \mathbb{Q} \rightarrow \mathbb{R}$ the p -adic norm given by $|0| = 0$ and $|p^t a/b|_p = 1/p^t$ if $a, b \in \mathbb{Z}$, $p \nmid ab$. The p -adic field $\mathbb{Q}_p$ is the completion of $(\mathbb{Q}, |\cdot|_p)$. Prove that for every prime p and every $a_1, \dots, a_n \in \mathbb{Q}_p$ we have

$$\max_{i,j} \min_{(k,l) \neq (i,j)} |(a_i - a_j) - (a_k - a_l)|_p \geq \frac{1}{4^{n-1}} \min_{i \neq j} |a_i - a_j|_p.$$

380. Let p be a prime and let $A \in M_n(\mathbb{Z})$ be a matrix such that $p \mid \text{tr } A^k$ for all integers $k \geq 1$.

- (i) Prove that if $n < p$ then there is some $m \geq 1$ such that $A^m \in pM_n(\mathbb{Z})$.
- (ii) Is the converse true?

381. Prove or disprove: for any ring A there exists a map $f : A \rightarrow Z(A)$ such that $f(1) = 1$ and $f(a+b) = f(a) + f(b)$ for all $a, b \in A$. (Here $Z(A)$ denotes the center of A .)

382. Let $a_1, b_1, \dots, a_s, b_s \in \mathbb{Z}_2$ and let $f : \mathbb{Z}_2^{2s} \rightarrow \mathbb{Z}_2$,

$$f(X_1, Y_1, \dots, X_s, Y_s) = \sum_{i=1}^s (a_i X_i^2 + X_i Y_i + b_i Y_i^2).$$

Determine $|f^{-1}(0)|$.

383. Let $C = \mathbb{R}^2 \times \mathbb{R}_{>0}$ be the parameter space of all plane circles. Let $n > 0$ and denote by H_k the subset of C^n parameterizing the configurations of n plane circles whose interiors contain a fixed point Q , such that any two circles intersect and no k circles pass through the same point. Show that H_4 is path connected.

384. Let $M \in M_2(\mathbb{C})$ be an invertible matrix and let $k > 1$ be an integer.

- (a) Show that if M is not scalar, then for any two matrices $A, B \in M_2(\mathbb{C})$ with $A^k = B^k = M$ there exists $E \in M_2(\mathbb{C})$ such that $A = B + E$.
- (b) Is the converse of (a) true?

385. Let $x \in (0, \pi)$ and $f(x) = \frac{1}{\tan x} - \frac{1}{x}$. Prove that $f^{(n)}(x) < 0$ for $n = 0, 1, 2, \dots$

386. Let $n \geq 1$ be an integer. Find the minimum of

$$f(\sigma) = \sum_{i \leq n/2} \sigma(i) + \sum_{i > n/2} \sigma^{-1}(i)$$

over all permutations $\sigma \in S_n$. Determine an explicit value of σ that realizes this minimum.

387. Let $a > 0$ and let $(a_n)_{n \geq 0}$ be defined by $a_0 = 0$ and $a_{n+1} = \sqrt{a + a_n}$ for all $n \geq 0$. Prove that the set of all n such that $a_n \in \mathbb{Q}$ is finite.

388. Let $a, b, c \in (0, 1)$ be real numbers. Prove the inequality

$$\sum_{\text{cyc}} \frac{1}{1-a^4} + \sum_{\text{cyc}} \frac{1}{1-a^2bc} \geq \sum_{\text{sym}} \frac{1}{1-a^3b}.$$

389. Let $\mathcal{F}$ be the real vector space of continuous functions $f : [0, 1] \rightarrow \mathbb{R}$. Consider on $\mathcal{F}$ the distance $d(f, g) = \int_0^1 |f(x) - g(x)| dx$.

- (a) Show that the intersection of any affine line directed by a nowhere zero function with any sphere consists of at most two points.
- (b) Does this property hold for every affine line in $\mathcal{F}$?

390. Let $a_0, a_1, \dots, a_n \in \mathbb{C}$ be pairwise different. Solve the linear system

$$\sum_{k=0}^n x_k a_k^j = 1 \quad (j = 0, 1, \dots, n).$$

391. Let x_n be the sequence defined by $x_1 = 1$, $x_{n+1} = p_{x_n}$, where p_n is the n -th prime number. Determine the asymptotic behavior of $\sqrt[n]{x_n}$, i.e., find a function f such that $\sqrt[n]{x_n} \sim f(n)$ as $n \rightarrow \infty$.

392. We consider on $\mathbb{R}^n$ the norm $\|x\| = \sum_{k=1}^n |x_k|$.

(a) Prove that if $n = 2$ then this norm has the property (M): for all $z, z', d \in \mathbb{R}^2$,

$$\|z\| > \|z - d\| \text{ and } \|z'\| > \|z' - d\| \implies \|z + z'\| > \|z + z' - d\|.$$

(b) Does the norm have property (M) if $n = 3$?

393. Let A, B, C, D be four distinct points in a plane Π , which are not the vertices of a parallelogram. Let H be one of the halfspaces bounded by Π .

(i) In H consider the semicircles of diameters AB and CD that are orthogonal on Π . Prove that in H there is exactly

We denote by $C(AB, CD)$ the semicircle from (i). Similarly define $C(AC, BD)$ and $C(AD, BC)$.

(ii) Prove that $C(AB, CD)$, $C(AC, BD)$, $C(AD, BC)$ pass through the same point.

(iii) Prove that $C(AB, CD)$, $C(AC, BD)$, $C(AD, BC)$ are orthogonal on each other.

394. Find all polynomials $P \in \mathbb{Z}[X]$ such that for any $a, b, c \in \mathbb{Z}$,

$$a^2 + b^2 + c^2 \mid P(a) + P(b) + P(c).$$

395. Let $z_1, z_2, \dots, z_n \geq 1$. Put $P = \prod_{i=1}^n z_i$, $P_i = \prod_{j \neq i} z_j$ ($1 \leq i \leq n$). Prove the inequality

$$\sum_{i=1}^n \frac{1}{1+z_i} + \frac{n(n-2)}{1+\sqrt[n]{P}} \geq (n-1) \sum_{i=1}^n \frac{1}{1+\sqrt[n]{P_i}}.$$

396. Let F be a field and let V be an F -vector space. Denote by $T(V), S(V), \Lambda(V)$ the tensor, symmetric and exterior algebras over V , respectively. Let $I_{S'}$ be the subgroup of $T(V)$ generated by

$$x_1 \otimes \dots \otimes x_n - x_{\sigma(1)} \otimes \dots \otimes x_{\sigma(n)} \quad (x_i \in V, \sigma \in A_n).$$

Then $I_{S'}$ is a homogeneous ideal of $T(V)$ and we set $S'(V) = T(V)/I_{S'}$. Hence $S'(V) = \bigoplus_{n \geq 0} S'^n(V)$ is a graded algebra; denote the product by $\odot$. Thus the image of $x_1 \otimes \dots \otimes x_n$ in $S'(V)$ is $x_1 \odot \dots \odot x_n$.

(i) For $n \geq 1$ let $\rho_{S^m, S^n} : S^m(V) \rightarrow S^n(V)$ be the linear map given by $x_1 \odot \dots \odot x_n \mapsto x_1 \cdots x_n$.

For $n \geq 2$ find a linear map $\rho_{\Lambda^n, S^n} : \Lambda^n(V) \rightarrow S^n(V)$ such that the short sequence

$$0 \rightarrow \Lambda^n(V) \xrightarrow{\rho_{\Lambda^n, S^n}} S^n(V) \xrightarrow{\rho_{S^m, S^n}} S^n(V) \rightarrow 0$$

is exact.

(ii) If $F = \mathbb{F}_2$, prove that for any $n \geq 1$ there is a linear map $\rho_{S^n, \Lambda^n} : S^n(V) \rightarrow \Lambda^n(V)$ with $x_1 \cdots x_n \mapsto x_1 \wedge \dots \wedge x_n$.

For $n = 2, 3$ find a linear map $\rho_{T^{n-1}, S^n} : T^{n-1}(V) \rightarrow S^n(V)$ such that the short sequence

$$0 \rightarrow T^{n-1}(V) \xrightarrow{\rho_{T^{n-1}, S^n}} S^n(V) \xrightarrow{\rho_{S^n, \Lambda^n}} \Lambda^n(V) \rightarrow 0$$

is exact.

397. Let $n \geq 1$ be an integer and let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a function with the property that the image under f of any sphere of codimension 1 is a sphere of codimension 1 of the same radius. Prove that f is an isometry.

398. Let $A \in M_n(\mathbb{Q})$ be an invertible matrix.

(a) Prove that if for every positive integer k there exists $B_k \in M_n(\mathbb{Q})$ such that $B_k^k = A$, then all eigenvalues of A are

(b) Is the converse of (a) true?

399. Let $n \geq 3$ and let $P = a_n X^n + \dots + a_0 \in \mathbb{R}[X]$ with $a_i > 0$ for all i such that all roots of P' are real. If $0 \leq a < b$ prove that

$$\frac{\int_a^b (P'(x))^{-1} dx}{\int_a^b (P''(x))^{-1} dx} \geq \frac{P'(b) - P'(a)}{P(b) - P(a)}.$$

400. Let $S(n) := \sum_{k=0}^n (-2)^k \binom{2n-k}{n-k}$. Prove that

$$4(n+1)S(n) + (n+2)S(n+2) = 0$$

and conclude that $S(n) = (-1)^n \frac{2^n}{n+1} \binom{2n}{n}$.

402. Let $u : [a, b] \rightarrow \mathbb{R}$ be twice differentiable with $u'(a) = u'(b) = 0$ and let $\lambda \in \mathbb{R}$.

(1) Prove that $u''(c) = \lambda u(c)u'(c)$ for some $c \in (a, b)$.

(2) If moreover $u''(a) = 0$, prove that $(d-a)u''(d) = u'(d)(1 + \lambda(d-a)u(d))$ for some $d \in (a, b)$.

403. A parabola $\mathcal{P}$ has focus F at distance d from the directrix Δ . Find the maximum length of an arc of $\mathcal{P}$ corresponding to a chord of length L .

404. Let $F : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ be a function satisfying:

1) $|F(x, y)| \geq |x| + |y| \forall x, y \in \mathbb{Z}$; 2) There exist $m, n \geq 1$ and matrices $A = (a_{ij}), B = (b_{ij}) \in M_{m,n}(\mathbb{Z})$ such that

$$F(x, y) = \max_{1 \leq i \leq m} \min_{1 \leq j \leq n} (a_{ij}x + b_{ij}y) \quad \forall x, y \in \mathbb{Z}.$$

Prove that either $F(x, y) \geq 0$ for all x, y or $F(x, y) \leq 0$ for all x, y . Give an example for each of these two cases.

405. Let $f \in C^2([0, 1])$ such that

$$f(0) + 2f\left(\frac{1}{p}\right) + 2f\left(\frac{2}{p}\right) + \dots + 2f\left(\frac{p-1}{p}\right) + f(1) = 0,$$

where $p \geq 2$ is an integer. Prove that

$$\left(\int_0^1 f(x) dx \right)^2 \leq \frac{1}{120p^4} \int_0^1 (f''(x))^2 dx.$$

Proposed by Cristian Chiser, Craiova, Romania.

406. Let d be a positive integer. Define a $2d \times 2d$ matrix $\mathbf{M}(d)$ with entries in $\{-1, 0, 1\}$ as follows: For $1 \leq a \leq 2d$ and $1 \leq b \leq d$

$$M_{a,b} = 1 \text{ if } a = 2b, \quad M_{a,b} = -1 \text{ if } a = 2b - 1, \quad M_{a,b} = 0 \text{ otherwise,}$$

and for $1 \leq a \leq 2d$ and $d+1 \leq b \leq 2d$ let $b' = b - d$ and set

$$M_{a,b} = 1 \text{ if } a = 2b', \quad M_{a,b} = -1 \text{ if } a = 2b' - 1, \quad M_{a,b} = 0 \text{ otherwise.}$$

Prove that $\det \mathbf{M}(d) = (-1)^d$.

Proposed by Ángel Plaza, Universidad de Las Palmas de Gran Canaria, Spain.

407. (i) Let i, n be two integers with $2 \leq i \leq n-2$ and let $x_k, y_k \in \mathbb{R}$ for $k = 1, \dots, n$ such that $x_1 \geq \dots \geq x_n$, $y_1 \geq \dots \geq y_n$. Prove that

$$n \left(\sum_{j=1}^{i-1} x_j y_j + x_i y_{i+1} + x_{i+1} y_i + \sum_{j=i+2}^n x_j y_j \right) \geq \sum_{j=1}^n x_j \sum_{j=1}^n y_j.$$

(ii) As an application, prove that if $a_k, b_k \in \mathbb{R}$ for $k = 1, 2, 3, 4$ such that $a_1 \geq a_2 \geq a_3 \geq a_4$ and $b_1 \geq b_2 \geq b_3 \geq b_4$ then

$$\frac{a_1 a_2 b_1 b_2 + a_2 a_3 b_2 b_3 + a_3 a_4 b_3 b_4 + a_4 a_1 b_4 b_1}{4} \geq \frac{a_1 a_2 + a_2 a_3 + a_3 a_4 + a_4 a_1}{4} \cdot \frac{b_1 b_2 + b_2 b_3 + b_3 b_4 + b_4 b_1}{4}.$$

Proposed by Ovidiu Pop, Satu Mare, Romania.

408. Let p be a prime and let $n \geq 2$ be an integer with $p \nmid n$. Prove that

$$\mathbb{Z}_p[X^n] \cap \mathbb{Z}_p[X^n + X^p] = \mathbb{Z}_p.$$

Proposed by Victor Alexandru, University of Bucharest, Romania.

409. Let n, k be two integers with $n \geq 3$ and $0 < k \leq n$. We have n distinct points on a circle. Determine how many ways we can choose out of these n points k mutually disjoint pairs of consecutive points.

Proposed by Ionel Popescu, Simon Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania.

410. Let $a, b \in (0, 1)$ and let $(a_n)_{n \geq 0}, (b_n)_{n \geq 0}$ be two sequences such that $a_0 = a, b_0 = b$ and for any $n \geq 0$ we have either

$$a_{n+1} = a_n^{\alpha_n}, b_{n+1} = b_n^{\alpha_n} \text{ for some } \alpha_n \geq 1$$

or

$$a_{n+1} = \lambda_n a_n + (1 - \lambda_n), b_{n+1} = \lambda_n b_n + (1 - \lambda_n) \text{ for some } \lambda_n \in [0, 1].$$

Prove the following. (i) (a_n) is convergent iff (b_n) is convergent. (ii) We have $\lim_{n \rightarrow \infty} a_n = 0$ or 1 iff $\lim_{n \rightarrow \infty} b_n = 0$ or 1 , accordingly. (iii) If (a_n) and (b_n) are divergent then $\lim_{n \rightarrow \infty} (a_n - b_n) = 0$.

Proposed by Liviu Paunescu and Constantin-Nicolae Beli, Simon Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania.

411. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \int_0^x \cos \frac{1}{t} \cos \frac{3}{t} \cos \frac{5}{t} \cos \frac{7}{t} dt.$$

Prove that f is well defined, differentiable and $f'(0) = \frac{1}{8}$.

Proposed by Eugen J. Ionascu, Department of Mathematics, Columbus State University, Columbus, Georgia, U.S.A.

412. Let $[A, B, C, D]$ be an equifacial tetrahedron with the lengths of the sides a, b, c and the length of the heights h . Prove that $h > \frac{1}{\sqrt{2}} \max\{a, b, c\}$ and

$$\frac{a^2 b^2 c^2}{(2h^2 - a^2)(2h^2 - b^2)(2h^2 - c^2)} \geq 27.$$

Proposed by Marius Olteanu, S.C. Hidroconstructia S.A., sucursala Muntenia, Ramnicu Valcea, Romania.

413. Prove that $\int_0^\infty \frac{\cos x - \cos(tx)}{x} dx = \ln |t|$ for any $t \neq 0$.

Proposed by George Stoica, Department of Mathematical Sciences, University of New Brunswick, Canada.

414. Does there exist a sequence $(a_n)_{n \geq 1}$ whose terms are greater or equal than 1 such that the following conditions are satisfied:

$$\prod_{k=1}^n a_k < n^n \text{ for all } n \geq 1,$$

and the sequence $(x_n)_{n \geq 1}$ given by $x_n = \sum_{i=1}^n \frac{1}{1+a_i}$ is bounded for all $n \geq 1$?

Proposed by Cezar Lupu, University of Pittsburgh, USA.

415. For any positive integer n we define the polynomial

$$P_n = \sum_{\substack{1 \leq k \leq n \\ (k, n) = 1}} X^k.$$

Prove that P_n is divisible by the cyclotomic polynomial Φ_n if and only if n is not squarefree.

Proposed by Filip-Andrei Chindea, student, University of Bucharest.

416. Let M_n be the set consisting of $\emptyset$ and all non-equivalent expressions in n variables $X_1, \dots, X_n$, which are sets, that can be obtained by using only $\cup$ and $\cap$. Two expressions $E_1, E_2 \in M_n$ are considered equivalent if $E_1(A_1, \dots, A_n) = E_2(A_1, \dots, A_n)$ for any sets $A_1, \dots, A_n$. E.g. $M_2 = \{\emptyset, X_1, X_2, X_1 \cup X_2, X_1 \cap X_2\}$. (i) Describe all elements of M_n . (ii) Prove that $2^{\lfloor n/2 \rfloor} \leq |M_n| \leq 2^{2^n}$.

Proposed by Constantin-Nicolae Beli, Simon Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania.

417. Calculate

$$\int_0^1 \int_0^1 \frac{\log(1+x) - \log(1+y)}{x-y} dx dy.$$

Proposed by Ovidiu Furdui, Technical University of Cluj-Napoca, Romania and Cornel Vălean, Teremia Mare, Timiș, Romania.

418. (i) Let $R = k[X, Y]/(XY^2)$, k a field. Denote by x and y the residue classes of X and Y modulo the ideal (XY^2) , respectively. Show that the elements x and $x(1+y)$ are associates, that is, $xR = x(1+y)R$, but there is no invertible element $u \in R$ such that $ux = x(1+y)$. (ii) Show that we cannot find such elements in $\mathbb{Z}/n\mathbb{Z}$.

Proposed by Cornel Băețica, Faculty of Mathematics and Informatics, University of Bucharest, Bucharest, Romania.

419. Suppose that $n \geq 1$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a function such that the image under f of the interior of any sphere S of codimension 1 is the interior of a sphere of codimension 1 of the same radius. Prove that f is an isometry.

Proposed by Marius Cavachi, Ovidius University of Constanța, Constanța, Romania.

420. Let $a, b, c, d \in \mathbb{R}$, $c, d \neq 0$, such that $\frac{a}{c} < \frac{b}{d}$. We consider the Maclaurin expansion $e^{\frac{a+b}{c+d}z} = \sum_{n \geq 0} a_n z^n$. (i) Find an exact formula as a finite sum for a_n . (ii) Determine the asymptotic behaviour of a_n as $n \rightarrow \infty$.
Proposed by Constantin-Nicolae Beli, Simion Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania.

421. (i) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function. Find the value of the limit

$$\lim_{n \rightarrow \infty} n^2 \iint_{x^2+y^2 \leq 1, 0 \leq z \leq 1} \left(\frac{\sqrt{x^2+y^2}+z}{2} \right)^n f(x, y, z) dx dy dz.$$

- (ii) Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}$ be a continuous function. Find the value of the limit

$$\lim_{n \rightarrow \infty} n^2 \iint_{x^2+y^2 \leq 1, z^2+t^2 \leq 1} \left(\frac{\sqrt{x^2+y^2} + \sqrt{z^2+t^2}}{2} \right)^n f(x, y, z, t) dx dy dz dt.$$

Proposed by Dumitru Popa, Ovidius University of Constanta, Constanta, Romania.

422. Find a sequence $(x_n)_{n \geq 1}$ of real numbers with the following properties: $x_n \searrow 0$, $\sqrt{n}/(x_1 + \dots + x_n) \rightarrow 0$ and $x_{[\sqrt{n}]} / x_n \rightarrow 1$, where $[\sqrt{n}]$ denotes the integer part of $\sqrt{n}$. Can you find a sequence with the above properties, in which $\sqrt{n}$ is replaced by $\ln n$?

Proposed by George Stoica, University of New Brunswick, Saint John, Canada.

423. Determine all differentiable functions $f : [0, \infty) \rightarrow \mathbb{R}$ with $f(0) = 0$ such that a) f' is strictly positive and increasing, b) $\int_0^x (f'(t))^2 dt \geq f(x + f(x)) - f(x)$ for all $x \in [0, \infty)$.

Proposed by Florin Stănescu, Șerban Cioculescu School, Găești, Dâmbovița, Romania.

424. Let $f, g \in \mathbb{C}[X]$ be monic polynomials of the same degree with the property that $|f(z)| = |g(z)| = 1$ for an infinity of values of $z \in \mathbb{C}$. Prove that $f = g$.

Proposed by Marius Cavachi, Ovidius University of Constanta, Constanta, Romania.

425. Let $n \geq 2$ and $a_1, \dots, a_n \geq 0$ be integers and let $b_1, \dots, b_n$ and λ be positive real numbers. Find the necessary and sufficient condition for the function

$$f(x_1, \dots, x_n) = \frac{x_1^{a_1} \dots x_n^{a_n}}{(|x_1|^{b_1} + \dots + |x_n|^{b_n})^\lambda}$$

to have a finite limit at $(0, \dots, 0)$.

Proposed by Dumitru Popa, Ovidius University of Constanta, Constanta, Romania.

426. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$, continuous in at least one point, and that satisfy the inequalities

$$f(x-1) \leq f(x) - 1, \quad f(x + \sqrt{2}) \leq f(x) + \sqrt{2} \quad \text{for } x \in \mathbb{R}.$$

Proposed by George Stoica, Department of Mathematical Sciences, University of New Brunswick, Canada.

427. Let F be a field of characteristic $\neq 2$, and let E/F be a finite multiquadratic extension so that $G := \text{Gal}(E/F) \cong \mathbb{Z}_2^n$. Let $a \in E^\times$ with the property that $a^{s-1} \in (E^\times)^2$ for all $s \in G$. Prove that there are $b_s \in E^\times$ with $s \in G$ such that $a^{s-1} = b_s^2$ for all $s \in G$ and

$$b_{stu}b_{st}^{-1}b_{su}^{-1}b_{tu}^{-1}b_sb_tb_u = b_u^{(s-1)(t-1)} \quad \forall s, t, u \in G.$$

How many $(b_s)_{s \in G} \in (E^\times)^G$ with the properties above exist?

Proposed by Constantin-Nicolae Beli, Simon Stoilow Institute of Mathematics of the Romanian Academy, Bucharest, Romania.

428. Show that $f(x) = x^{4n} - 3x^{3n} + 4x^{2n} - 2x^n + 1$ is irreducible in $\mathbb{Z}[x]$ for all integers $n \geq 1$.
Proposed by Cornel Băețica, Faculty of Mathematics and Informatics, University of Bucharest, Bucharest, Romania.

429. Let f be a C^1 -class real valued function on $[0, 1]$, infinitely differentiable at $x = 0$. If $f^{(n)}(0) = 0$ for every $n \in \mathbb{N}$ and, for some $C > 0$ one has $|xf'(x)| \leq C|f(x)|$ for every $x \in [0, 1]$, then $f(x) = 0$ for every $x \in [0, 1]$.

Proposed by George Stoica, Department of Mathematical Sciences, University of New Brunswick, Canada.

430. Suppose that $r \in \mathbb{Q}$ and $m \in \mathbb{Z}$, $m > 0$, such that $\cos^m r\pi \in \mathbb{Q}$. Determine all possible values of $\cos r\pi$.
Proposed by Marius Cavachi, Ovidius University, Constanta, Romania.

431. Let $f : [0, 1]^2 \rightarrow \mathbb{R}$ be such that $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} : [0, 1]^2 \rightarrow \mathbb{R}$ are continuous. Find the value of the limit

$$\lim_{n \rightarrow \infty} n \left(n^2 \iint_{[0, 1]^2} x^n y^n f(x, y) \, dx dy - f(1, 1) \right).$$

Proposed by Dumitru Popa, Ovidius University Constanta, Romania.

432. Let $k \geq 1$ be an integer. Find all $\alpha \in \mathbb{R}$ with the property that there is a sequence $(a_n)_{n \geq 1}$ such that $a_1 + \dots + a_{n+k} < \alpha a_n$ for all $n \geq 1$.

Proposed by Constantin-Nicolae Beli, IMAR, Bucuresti, Romania.

433. Prove that

$$\sum_{k=1}^{\infty} k(k+1 - \zeta(2) - \zeta(3) - \dots - \zeta(k+1)) = \zeta(3),$$

where ζ denotes the Riemann zeta function.

Proposed by Ovidiu Furdui, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

434. If $-1 \leq \alpha < 0$, show that the function

$$f(x) = (x+4)^\alpha - 3(x+6)^\alpha + x(x+3)^\alpha - x(x+5)^\alpha$$

is strictly increasing for $x \geq 0$.

Proposed by Ioan Tomescu, Faculty of Mathematics and Informatics, University of Bucharest, Romania.

435. Let $f(x) = \frac{e^x}{(a+be^x)^2}$, where a, b are two integers with $a+b=1$. (i) Prove that $f^{(k)}(0) \in \mathbb{Z}$ for any $k \geq 0$.
(ii) Prove that if p is a prime and $\alpha \geq 1$ then for any integers $k, l \geq 0$ with $k \equiv l \pmod{p^{\alpha-1}(p-1)}$ we have $f^{(k)}(0) \equiv f^{(l)}(0) \pmod{p^\alpha}$.

Proposed by Constantin-Nicolae Beli, IMAR, Bucuresti, Romania.

436. Calculate

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln \sqrt{n(n+1)} - \gamma \right),$$

where γ denotes the Euler–Mascheroni constant.

Proposed by Ovidiu Furdui, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

437. Let $f : [0, 1] \rightarrow \mathbb{R}$ with $f(0) \in \mathbb{Q}$. Suppose that for every $x \in [0, 1]$ there exists $\epsilon_x > 0$ such that $f(x) - f(y) \in \mathbb{Q}$ for $y \in [0, 1]$ and $|x - y| < \epsilon_x$. Prove that $f(x) \in \mathbb{Q}$ for every $x \in [0, 1]$.

Proposed by George Stoica, Department of Mathematical Sciences, University of New Brunswick, Canada.

438. Let $f_1, \dots, f_n$ be polynomials with real coefficients with positive leading coefficients. Suppose that there is $M \in \mathbb{R}$ and $f \in \mathbb{R}[X]$ such that $f_i(x) > 0$ for $1 \leq i \leq n$ and $\sqrt{f_1(x)} + \dots + \sqrt{f_n(x)} = f(x)$ for all $x > M$. Prove that for any $1 \leq i \leq n$ we have $f_i = g_i^2$ for some $g_i \in \mathbb{R}[X]$.

Proposed by Marius Cavachi, Ovidius University, Constanta, Romania.

447. Let $A \in \mathcal{M}_2(\mathbb{Z})$. Prove that $e^A \in \mathcal{M}_2(\mathbb{Z})$ if and only if $A^2 = O_2$.

Proposed by Ovidiu Furdui, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

448. Let $f : [0, \infty) \rightarrow (0, \infty)$ be a continuous function and let $F(x) = \int_0^x f(t)dt$. Let $k > 0$. (i) Prove that if $\alpha < ke$ then there are $\epsilon, A > 0$ with the property that

$$\int_0^{2A} e^{-\epsilon(x-A)^2} (F(x+k) - \alpha f(x)) dx > 0.$$

Moreover ϵ, A are independent of f . (ii) Give an example of f such that $F(x+k) < kef(x)$ for all $x > 0$. Conclude that if $k > 0$ and $\alpha \in \mathbb{R}$ then there is $f : [0, \infty) \rightarrow (0, \infty)$ such that $\int_0^{x+k} f(t)dt < \alpha f(x)$ for all $x > 0$ if and only if $\alpha \geq ke$.

Proposed by Constantin-Nicolae Belu, IMAR, Bucuresti, Romania.

449. Let m, n be positive integers and let $f, g_1, \dots, g_{n+1} \in \mathbb{Z}[X]$, where f is monic with $\deg f = n$, such that all roots of f are real and $|g_i(\alpha)| < m$ for $1 \leq i \leq n+1$ and for every root α of f . Show that there exist $n+1$ integers, not all zero, each of absolute value at most $(2m(n+1))^n$, such that every root of f is also a root of $\sum_{i=1}^{n+1} a_i g_i$.

Proposed by Marius Cavachi, Ovidius University, Constanta, Romania.

450. Find all polynomials $P \in \mathbb{R}[x]$ with the property

$$P(\sin x) + P(\cos x) = 1 \quad \forall x \in \mathbb{R}.$$

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

451. Let V be the linear space of the polynomial functions with real coefficients, defined on $[0, 1]$. Denote $e_n(x) = x^n$, $x \in [0, 1]$, $n = 0, 1, 2, \dots$. Let $T : V \rightarrow V$ be a linear operator such that: (1) $Te_0 = e_0$, $Te_1 = Te_2 = e_1$; (2) If $p \in V$, $p(x) \geq 0$ for all $x \in [0, 1]$, then $(Tp)(x) \geq 0$ for all $x \in [0, 1]$. Prove that T is a projection, and find the null space $\text{Ker } T$ and the range $\text{Im } T$.

Proposed by Ioan Raşa, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

452. Let $(T_n)_{n \in \mathbb{N}}$ be the sequence of Chebyshev's polynomials, which on the interval $[-1, 1]$ are defined by $T_n(x) = \cos(n \arccos x)$, $n \in \mathbb{N}$, and let $(F_n)_{n \geq 0}$ be the Fibonacci sequence defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for all $n \geq 1$. Prove that

$$T_n\left(-\frac{3}{2}\right) = 1 + (-1)^n \frac{5}{2} F_n^2 \quad \forall n \in \mathbb{N}.$$

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

453. Let $f, g \in \mathbb{Q}[X]$ be monic and irreducible polynomials over $\mathbb{Q}$ with the property that there are $\alpha, \beta \in \mathbb{C}$ with $\alpha + \beta \in \mathbb{Q}$ and $f(\alpha) = g(\beta) = 0$. Prove that the polynomial $f^2 - g^2$ has a root in $\mathbb{Q}$.

Proposed by Bogdan Moldovan, student, Babeş-Bolyai University, Cluj-Napoca, Romania.

454. Let $A \in M_3(\mathbb{Z})$ with $A^{30} - 2A^{25} = A^5 - 2I_3$. Prove that $\text{tr}(A) \not\equiv 2 \pmod{5}$.

Proposed by Luigi-Ionuţ Catana, student, University of Bucharest, Bucureşti, Romania.

455. Let $n \geq 2$ be an integer. Determine the largest number of real solutions the equation $a_1\sqrt{x+b_1} + \dots + a_n\sqrt{x+b_n} = 0$ can have. Here $a_1, \dots, a_n$ are real numbers, not all zero, and $b_1, \dots, b_n$ are mutually distinct numbers.

Proposed by Marius Cavachi, Ovidius University, Constanta, Romania.

456. Let f, g, h be non-negative continuous functions on $[0, 1]$ satisfying the inequality $f(tx + (1-t)y) \geq g^t(x)h^{1-t}(y)$ for all $x, y \in [0, 1]$ and some (fixed) $t \in (0, 1)$. If, in addition, we have $\int_0^1 g(x)dx = \int_0^1 h(x)dx = 1$, prove that $\int_0^1 f(x)dx \geq 1$.

Proposed by George Stoica, University of New Brunswick, Saint John, New Brunswick, Canada.

457. Let $A, B \in \mathcal{M}_n(\mathbb{C})$ so that $A^2 + B^2 + A - B = 2(AB + I_n)$. Prove the following equalities:

$$\text{Tr}((A-B)(A-B+I_n)) = 2n, \quad \det((A-B)(A-B+I_n)) = 2^n.$$

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

458. For a continuous and non-negative function f on $[0, 1]$ we define the Hausdorff moments

$$\mu_n := \int_0^1 x^n f(x) dx, \quad n = 0, 1, 2, \dots$$

Prove that $\mu_{n+2}\mu_0^2 + \mu_n\mu_1^2 \geq 2\mu_{n+1}\mu_1\mu_0$ for $n = 0, 1, 2, \dots$ *Proposed by Cezar Lupu, University of Pittsburgh, Pittsburgh, PA, USA.*

459. The faces of an icosahedron are colored with blue or white such that a blue face cannot be adjacent to more than two other blue faces. What is the largest number of blue faces that can be obtained following this rule? (Two faces are considered adjacent if they share an edge.)
Proposed by Eugen J. Ionascu, Columbus State University, Columbus, GA, USA.

460. Let X be a set with at least two elements, and fix $a, b \in X$, $a \neq b$. We define the function $f : X^3 \rightarrow X$ by

$$f(x, y, z) = \begin{cases} a, & \text{if } x = y = z = a, \\ b, & \text{otherwise.} \end{cases}$$

Is there a binary operation $*$: $X^2 \rightarrow X$ such that $f(x, y, z) = (x * y) * z$ for all $x, y, z \in X$?

Proposed by George Stoica, University of New Brunswick, Saint John, New Brunswick, Canada.

461. Let $A, B \in \mathcal{M}_n(\mathbb{R})$ so that $A^2 = A$, $B^2 = B$, and $\det(2A + B) = 0$. Prove that $\det(A + 2B) = 0$.
Proposed by Vasile Pop, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

462. If $f : [0, 1] \rightarrow \mathbb{R}$ is a convex function with $f(0) = 0$ then prove that

$$\frac{1}{6} \left(\int_{1/2}^1 f(x) dx - \int_0^{1/2} f(x) dx \right) \geq \int_0^{1/2} x f(x) dx.$$

Proposed by Florin Stănescu, Șerban Cioculescu School, Găești, Dâmbovița, Romania.

463. Prove that there exists $n_0 \in \mathbb{N}$ such that $\forall n \geq n_0$ the equation

$$\frac{1}{1+x} + \frac{1}{2+x} + \dots + \frac{1}{n+x} = \ln n$$

has a unique solution in the interval $(0, \infty)$, denoted by x_n , and that

$$\lim_{n \rightarrow \infty} x_n = a,$$

where $a \in (0, 1)$ is the unique solution in $(0, 1)$ of the equation $x \sum_{i=1}^{\infty} \frac{1}{i(i+x)} = \gamma$, where γ is the Euler constant.

Proposed by Dumitru Popa, Ovidius University, Constanța, Romania.

464. Prove that the only twice differentiable functions $f : (0, \infty)^2 \rightarrow \mathbb{R}$ satisfying

$$x^2 \frac{\partial^2 f}{\partial x^2}(x, y) + 2xy \frac{\partial^2 f}{\partial x \partial y}(x, y) + y^2 \frac{\partial^2 f}{\partial y^2}(x, y) + \frac{1}{4} f(x, y) = 0$$

are $f(x, y) = C(x, y) + D(x, y) \log x$, where C and D are twice differentiable, homogeneous functions of degree $1/2$.

Proposed by George Stoica, Saint John, New Brunswick, Canada.

465. Let $A \in M_{m,n}(\mathbb{C})$, $B \in M_{n,m}(\mathbb{C})$. Prove that

$$\text{rank}(I_m - AB) - \text{rank}(I_n - BA) = m - n.$$

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

466. Let $f : [1, \infty) \rightarrow \mathbb{R}$ be differentiable with $\lim_{x \rightarrow \infty} f(x) = 0$. Suppose that f' is strictly monotone, does not vanish anywhere and has the property $\lim_{n \rightarrow \infty} \frac{f'(n+1)}{f'(n)} = 1$. Prove that the limit

$$\gamma_f := \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n f(k) - \int_1^n f(x) dx \right)$$

exists and is finite. Calculate $\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n f(k) - \int_1^n f(x) dx - \gamma_f}{f(n)}$.

Proposed by Mircea Ivan, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

467. Let $a, b, c, d, n \geq 0$ be some integers with $a + c, b + d \geq n$. Then

$$\sum_{k+l=n} k! l! \binom{a}{l} \binom{b}{l} \binom{d}{k} \binom{a+c-l}{k} = \sum_{k+l=n} k! l! \binom{a}{l} \binom{c}{k} \binom{d}{k} \binom{b+d-k}{l}.$$

Proposed by Constantin-Nicolae Belî, IMAR, Bucuresti, Romania.

468. Find all functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfying

$$f(xu + yv, -xv + yu) = f(x, y)f(u, v) \quad \forall x, y, u, v \in \mathbb{R}.$$

Which ones are continuous?

Proposed by George Stoica, Saint John, New Brunswick, Canada.

469. Let $A, B \in M_3(\mathbb{R})$ such that $A^2 + BA + B^2 = AB$ and $S(A) \leq 0$, where $S(A)$ is the sum of minors corresponding to the entries on the diagonal of A . Prove that $(AB)^2 = A^2B^2$.

Proposed by Florin Stănescu, Șerban Cioculescu Highschool, Găești, Dâmbovița, Romania.

470. Let $a \in \mathbb{R}$ and let $b, c \in \mathbb{R}$ with $bc > 0$. Calculate

$$\lim_{n \rightarrow \infty} \begin{pmatrix} 1 & b/n \\ c/n & 1 + a/n^2 \end{pmatrix}^n.$$

Proposed by Ovidiu Furdui, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

471. Let R be a commutative ring of characteristic 2, i.e., where $2 = 0$. In the ring $R[[X]]$ we consider the formal power series $f(X) = \sum_{n \geq 0} X^{2^n - 1}$. Prove that for any $\alpha, \beta \in R$ the formal power series $f(\alpha^2 X)f(\beta^2 X)f((\alpha + \beta)^2 X)$ is a square in $R[[X]]$.

Proposed by Constantin-Nicolae Belî, IMAR, Bucuresti, Romania.

472. Let $a, b, c \in [0, \frac{\pi}{2}]$ such that $a + b + c = \pi$. Prove the following inequality:

$$\sin a + \sin b + \sin c \geq 2 + 4 \left| \sin \frac{a-b}{2} \sin \frac{b-c}{2} \sin \frac{c-a}{2} \right|.$$

Proposed by Leonard Giugiac, National College Traian, Drobeta Turnu Severin, Romania and Jiahao He, South China University of Technology, People's Republic of China.

473. (Corrected) Let $e_1, \dots, e_n$ be the elementary symmetric polynomials in the variables $X_1, \dots, X_n$

$$e_k(X_1, \dots, X_n) = \sum_{1 \leq i_1 < \dots < i_k \leq n} X_{i_1} \cdots X_{i_k},$$

and let M be the ideal generated by $e_1, \dots, e_n$ in $\mathbb{R}[X_1, \dots, X_n]$. Then every monomial $X_1^{m_1} \cdots X_n^{m_n}$ with degree $m = m_1 + \dots + m_n$ strictly greater than $\binom{n}{2}$ belongs to M . On the other hand, there exists a monomial of degree $\binom{n}{2}$ which does not belong to M .

Proposed by George Stoica, New Brunswick, Canada.

474. Calculate

$$\sum_{n=1}^{\infty} (2n-1) \left[\left(\frac{1}{n^2} + \frac{1}{(n+1)^2} + \dots \right)^2 - \frac{1}{n^2} \right].$$

Proposed by Ovidiu Furdui and Alina Sintamarian, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

475. We say that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ has the property (P) if it is continuous and

$$2f(f(x)) = 3f(x) - x \quad \text{for all } x \in \mathbb{R}.$$

a) Prove that if f has property (P) then $M = \{x \in \mathbb{R} : f(x) = x\}$ is a nonempty interval. b) Find all functions with property (P). *Proposed by Dan Moldovan and Bogdan Moldovan, Cluj-Napoca, Romania.*

476. Calculate the integral

$$\int_0^{\infty} \frac{\arctan x}{\sqrt{x^4 + 1}} dx.$$

Proposed by Vasile Mircea Popa, Lucian Blaga University, Sibiu, Romania.

477. For every complex matrix A we denote by A^* its adjoint, i.e., the transposed of its conjugate, $A^* = \bar{A}^T$. If A is square and $A = A^*$ we say that A is self-adjoint (or Hermitian). In this case for every complex vector x we have $x^*Ax \in \mathbb{R}$. If A, B are self-adjoint we say that $A \geq B$ if $x^*Ax \geq x^*Bx$ for every complex vector x . For a complex matrix A we denote $|A|^2 = AA^*$. Note that $|A|^2$ is self-adjoint and ≥ 0 . Let A be a square matrix with complex coefficients and I the identity matrix of the same order. Then the following statements are equivalent: (i) $|I + zA|^2 = |I - zA|^2$ for all $z \in \mathbb{C}$; (ii) $|I + zA|^2 \geq I$ for all $z \in \mathbb{C}$; (iii) $A = 0$. Are these statements still equivalent if we replace "complex" by "real" throughout?
Proposed by George Stoica, New Brunswick, Canada.

478. Determine the largest positive constant k such that for every $a, b, c \geq 0$ with $a^2 + b^2 + c^2 = 3$ we have

$$(a + b + c)^2 + k|(a - b)(b - c)(c - a)| \leq 9.$$

Proposed by Leonard Giugiuc, National College Traian, Drobeta Turnu Severin, Romania.

479. Let p be an odd prime number and $A \in \mathcal{M}_p(\mathbb{Q})$ a matrix such that $\det(A^p + I_p) = 0$ and $\det(A + I_p) \neq 0$. Prove that: a) $\text{Tr}(A)$ is an eigenvalue of $A + I_p$; b) $\det(A + I_p) - \det(A - I_p) = (p - 1)\text{Tr}(A) + 2$.
Proposed by Vlad Mihalý, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

480. Let k, n be natural numbers, $x_1, x_2, \dots, x_k$ be distinct complex numbers and the matrix $A \in \mathcal{M}_n(\mathbb{C})$ such that $(A - x_1 I_n)(A - x_2 I_n) \cdots (A - x_k I_n) = O_n$. Prove that

$$\text{rank}(A - x_1 I_n) + \text{rank}(A - x_2 I_n) + \cdots + \text{rank}(A - x_k I_n) = n(k - 1).$$

Proposed by Dan Moldovan and Vasile Pop, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

482. Let $x \in \mathbb{R}$. Prove the series

$$\sum_{n=0}^{\infty} 3^n \left(\sin x - x + \frac{1}{3!}x^3 - \cdots - \frac{\sin \frac{n\pi}{2}}{n!}x^n \right)$$

converges absolutely and calculate its sum.

Proposed by Ovidiu Furdui, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

483. Suppose that $0 < a_1 \leq a_2 \leq \cdots$ and $\sum_{n=1}^{\infty} 1/a_n < \infty$. Let $A := \{n \in \mathbb{N}^* : a_n < n \log n\}$. Prove that A has logarithmic density 0, that is, $\lim_{x \rightarrow \infty} \frac{1}{\log x} \sum_{\substack{k \leq x \\ k \in A}} \frac{1}{k} = 0$.

Proposed by George Stoica, New Brunswick, Canada.

484. Prove that for a continuous nonconstant function $f : \mathbb{R} \rightarrow \mathbb{R}$, the following conditions are equivalent: (1) $f(x) - f(y) \in \mathbb{Q}$ for all $x, y \in \mathbb{R}$ such that $x - y \in \mathbb{Q}$; (2) $f(x) - f(y) \in \mathbb{R} \setminus \mathbb{Q}$ for all $x, y \in \mathbb{R}$ such that $x - y \in \mathbb{R} \setminus \mathbb{Q}$; (3) there exist $a \in \mathbb{Q}^*$ and $b \in \mathbb{R}$ such that $f(x) = ax + b$ for all $x \in \mathbb{R}$.
Proposed by Vasile Pop, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

485. Assume that ABC is a triangle with $a \geq b \geq c$, where the angle A has a fixed value. Denote by Σ the sum

$$\sqrt{\frac{b+c-a}{a}} + \sqrt{\frac{c+a-b}{b}} + \sqrt{\frac{a+b-c}{c}}.$$

Then the only possible values of A are $\pi/3 \leq A < \pi$ and we have: (i) The smallest possible value Σ is

$$\frac{4 \sin \frac{A}{2} + \sqrt{2(1 - \sin \frac{A}{2})}}{\sqrt{2 \sin \frac{A}{2}}}.$$

(ii) If $\pi/3 \leq A < \pi/2$ then the largest possible value of Σ is

$$\frac{4 \cos A + \sqrt{2(1 - \cos A)}}{\sqrt{2 \cos A}}.$$

(iii) If $\pi/2 \leq A < \pi$ then there is no finite upper bound for Σ .

Proposed by Leonard Giugiuc, National College Traian, Drobeta Turnu Severin, Romania.

486. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which satisfy the equation

$$f(-x) = x + \int_0^x \sin t f(x-t) dt \quad \forall x \in \mathbb{R}.$$

Proposed by Ovidiu Furdui and Alina Sintamarian, Technical University of Cluj-Napoca, Cluj-Napoca, Romania.

487. We consider the complex matrices A and B of dimensions $m \times n$ and $n \times m$, where $m > n \geq 2$. Let $C = AB$. If 0 is an eigenvalue of C of order $m - n$ and $C^{k+1} = \lambda C^k$ for some $k \geq 1$ and $\lambda \in \mathbb{C} \setminus \{0\}$, then determine BA .

Proposed by Leonard Giugiu, National College Traian, Drobeta Turnu Severin, and Costel Balcau, University of Pitesti, Romania.

488. Let $0 \leq a < b$ and $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function with $f(a) = f(b)$. Then there exist $\alpha, \beta \in [a, b]$ such that $f(\alpha) = f(\beta)$ and $\alpha/\beta \notin \mathbb{Q}$.

Proposed by George Stoica, New Brunswick, Canada.

489. Let $m \leq n$ be positive integers. For $A \in \mathcal{M}_{m,n}(\mathbb{C})$ and $B \in \mathcal{M}_{n,m}(\mathbb{C})$ define the functions

$$f_{A,B} : \mathcal{M}_n(\mathbb{C}) \rightarrow \mathcal{M}_m(\mathbb{C}), \quad f_{A,B}(X) = AXB, \quad f_{B,A} : \mathcal{M}_m(\mathbb{C}) \rightarrow \mathcal{M}_n(\mathbb{C}), \quad f_{B,A}(Y) = BYA.$$

Prove that $f_{A,B}$ is surjective (onto) if and only if $f_{B,A}$ is injective (one-to-one).

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Romania.

490. Let $n \in \mathbb{N}^*$. Calculate

$$\int_0^1 \left(\frac{\ln(1-x) + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n}}{x} \right)^2 dx.$$

Proposed by Ovidiu Furdui and Alina Sintamarian, Technical University of Cluj-Napoca, Romania.

491. If the arithmetic mean of $a, b, c, d \geq 0$ is 1, then their quadratic mean $q = \sqrt{\frac{a^2+b^2+c^2+d^2}{4}}$ takes values in the interval $[1, 2]$. If $q \in [1, 2]$ then we denote by $M = M_q$ the largest possible value of the geometric mean of four numbers $a, b, c, d \geq 0$ with the arithmetic mean 1 and the quadratic mean q . Determine M in terms of q and prove that $M + q \geq 2$.

Proposed by Leonard Giugiu, Traian National College, Drobeta Turnu Severin, Romania and Alexander Bogomolny, New Jersey, USA.

492. Let V be a vector space over $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ and $f : V \rightarrow \mathbb{R} \cup \{\infty\}$ satisfying $f(x) = \infty$ iff $x = 0$ and

$$f(x+y) \geq \min\{f(x), f(y)\} \quad \forall x, y \in V.$$

For every $c \in V$ define $g_c : V \rightarrow \mathbb{R} \cup \{\infty\}$ by $g_c(x) = f(x) + f(x+c)$. Prove that g_c satisfies the same inequality as f . Equivalently, if $x, y, z, t \in V$ with $x+y = z+t$ then

$$f(x+z) + f(x+t) \geq \min\{f(x) + f(y), f(z) + f(t)\}.$$

For any $a, b \in V$ define $h_{a,b} : V \rightarrow \mathbb{R} \cup \{\infty\}$ by $h_{a,b}(x) = f(x) + f(x+a) + f(x+b)$. (ii) If $x, y, a, b \in V$ such that $f(x) \leq f(y)$ prove that

$$h_{x,x+a+b}(y) \geq \min\{h_{a,b}(x), h_{a,b}(y)\}.$$

Let $k : V^2 \rightarrow \mathbb{R} \cup \{\infty\}$, $k(x, y) = f(x) + f(y) + f(x+y)$. (iii) If $a, b, x, y \in V$ prove that

$$h_{a,b}(x+y) \geq \min\{h_{a,b}(x), h_{a,b}(y), k(x, y)\}$$

and

$$k(x, y) \geq \min\{h_{a,b}(x), h_{a,b}(y), h_{a,b}(x+y)\}.$$

Conclude that none of the four numbers $h_{a,b}(x), h_{a,b}(y), h_{a,b}(x+y), k(x, y)$ is strictly smaller than all remaining three numbers. (iv) If $a, b, x, y, z \in V$ prove that

$$\max\{h_{y,z}(x), h_{z,x}(y), h_{x,y}(z)\} \geq \min\{h_{a,b}(x), h_{a,b}(y), h_{a,b}(z)\}.$$

Proposed by Constantin-Nicolae Beli, IMAR, București, Romania.

493. (a) Calculate

$$\lim_{n \rightarrow \infty} n \int_0^\infty \frac{\sin x}{e^{(n+1)x} - e^{nx}} dx.$$

(b) Let $k > -1$ be a real number. Calculate

$$\lim_{n \rightarrow \infty} n^{k+1} \int_0^\infty \frac{x^k \sin x}{e^{(n+1)x} - e^{nx}} dx.$$

Proposed by Ovidiu Furdui and Alina Sintamarian, Technical University of Cluj-Napoca, Romania.

494. Let $n \geq 3$ and let $a_1, \dots, a_n$ be nonnegative real numbers such that $a_1^2 + \dots + a_n^2 = n - 1$. (i) Prove that $a_1 + \dots + a_n - a_1 \cdots a_n \leq n - 1$. (ii) Prove that if $k < 1$ then the inequality $a_1 + \dots + a_n - k a_1 \cdots a_n \leq n - 1$ is not always true.

Proposed by Leonard Giugiu, Traian National College, Drobeta Turnu Severin, Romania, Qing Song, Beihang University Library and Yongxi Wang, Shanxi University Affiliated High School, People's Republic of China.

495. Let $n \geq 2$ and $A, B \in \mathcal{M}_n(\mathbb{C})$ such that

$$AB - BA = c(A - B)$$

for some $c \in \mathbb{C}^*$. a) For $n = 2$, give an example of distinct matrices A and B that satisfy the above condition. b) Prove that A and B have the same eigenvalues.

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Romania, and Mihai Opincariu, Avram Iancu National College, Brad, Romania.

496. Calculate the integral:

$$\int_0^\infty \frac{\arctan x}{\sqrt{x^4 + x^2 + 1}} dx.$$

Proposed by Vasile Mircea Popa, Lucian Blaga University, Sibiu, Romania.

497. Let $n \geq 4$ and let $a_1, \dots, a_n$ be nonzero real numbers such that $\frac{1}{a_1} + \dots + \frac{1}{a_n} = 0$. Prove that

$$\left(\frac{1}{a_1^2} + \dots + \frac{1}{a_n^2} \right) \sum_{1 \leq i < j \leq n} (a_i - a_j)^2 \geq n^3.$$

When does equality hold?

Proposed by Leonard Giugiu, Traian National College, Drobeta Turnu Severin, Romania.

498. Let $A, B \in M_n(\mathbb{C})$ be two matrices such that

$$A^2 - B^2 - I_n = \frac{1}{3}(AB - BA).$$

Prove that: (i) $\det(A^2 - B^2) = \det(A - B) \det(A + B) = 1$. (ii) $(AB - BA)^n = 0$.

Proposed by Florin Stănescu, Șerban Cioculescu School, Găești, Dâmbovița, Romania.

499. Let $a, b \geq 0$. Calculate

$$\lim_{n \rightarrow \infty} \sqrt{n} \int_0^{\frac{\pi}{2}} \sqrt{a \sin^{2n} x + b \cos^{2n} x} dx.$$

Proposed by Ovidiu Furdui and Alina Sintămărian, Technical University of Cluj-Napoca, Romania.

500. Let C be a simplex in $\mathbb{R}^n$ with vertices $A_1, \dots, A_{n+1}$ and let M be a point in the interior of C . For every $1 \leq i < j \leq n+1$ denote by $A_{i,j}$ the point where the hyperplane generated by M and $A_1, \dots, \hat{A}_i, \dots, \hat{A}_j, \dots, A_n$ intersects the edge $A_i A_j$ of C . Denote by D the convex hull of $\{A_{i,j} : 1 \leq i < j \leq n+1\}$. Prove that $\text{vol } D \leq (1 - \frac{n+1}{2^n}) \text{vol } C$, and equality holds iff M is the centroid of C .

Proposed by Leonard Giugiu, National College Traian, Drobeta Turnu Severin, Costel Balcau, University of Pitești, and Constantin-Nicolae Beli, IMAR, București, Romania.

501. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function. Then

$$f(x+y) - f(x) \geq y f'(x) \quad \forall x, y \in \mathbb{R}$$

if and only if

$$n(f(x+1/n) - f(x)) \geq f'(x) \quad \forall x \in \mathbb{R} \text{ and every positive integer } n.$$

Proposed by Florin Stănescu, Șerban Cioculescu School, Găești, Dâmbovița, Romania.

502. Let $m \geq 0$ be an integer. Evaluate the series

$$\sum_{k=1}^{\infty} \frac{(x^m \log x)^{(k+m)}}{k!}, \quad x > 1,$$

where $f^{(i)}$ is the derivative of order i of f .

Proposed by Mircea Ivan, Technical University of Cluj-Napoca, Romania.

503. The Poincaré half-space model of the non-Euclidean n -dimensional space is the upper half-space $\mathbb{H}_n = \{(x, y) \mid x \in \mathbb{R}^{n-1}, y > 0\}$. We regard an element $x \in \mathbb{R}^{n-1}$ as a column vector. The group of positively oriented isometries of $\mathbb{H}_n$ consists of the functions $f_{\alpha, A, a} : \mathbb{H}_n \rightarrow \mathbb{H}_n$ with $\alpha > 0$, $A \in O^+(n-1)$, $a \in \mathbb{R}^{n-1}$, given by $(x, y) \mapsto \alpha(Ax + a, y)$, and the functions $g_{\alpha, A, r, a} : \mathbb{H}_n \rightarrow \mathbb{H}_n$ with $\alpha > 0$, $A \in O^-(n-1)$, $r, a \in \mathbb{R}^{n-1}$, given by $(x, y) \mapsto \alpha\left(\frac{A(x-r)}{|x-r|^2+y^2} + b, \frac{y}{|x-r|^2+y^2}\right)$. Give a direct proof that if G is the set of all $f_{\alpha, A, a}$ and all $g_{\alpha, A, r, a}$ then $(G, \circ)$ is a group.

Proposed by Constantin-Nicolae Beli, IMAR, București, Romania.

504. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with f' continuous on $[0, 1]$, such that $|f'(x)| \leq 1$ for all $x \in [0, 1]$. Prove that if $2\left|\int_0^1 f(x)dx\right| \leq 1$ then $(n+2)\left|\int_0^1 x^n f(x)dx\right| \leq 1$ for all $n \geq 1$.

Proposed by Florin Stănescu, Șerban Cioculescu School, Găești, Dâmbovița, Romania.

505. Let $n, p, q \in \mathbb{N}$ such that $1 \leq q < p < n$. If there exists $A \in \mathcal{M}_n(\{0, 1\})$ such that AA^t has all diagonal elements equal to p and all off-diagonal elements equal to q , prove that: a) $p(p-1) = q(n-1)$; b) $AA^t = A^tA$; c) if n is even, then $p-q$ is a perfect square.

Proposed by Vasile Pop and Mircea Rus, Technical University of Cluj-Napoca, Romania.

506. Let $N > 1$ be a squarefree integer. For every integer k denote $q_N(k) = \gcd(N, k)$. Prove that there is a finite subset S of the unit circle such that for every polynomial $f = \sum_{k=0}^n a_k X^k \in \mathbb{C}[X]$ we have

$$\mu(N) \sum_{k=0}^n \mu(q_N(k)) \phi(q_N(k)) a_k = \sum_{\zeta \in S} f(\zeta).$$

(Here μ and ϕ denote the Möbius function and Euler's totient function, respectively.)

Proposed by Marian Tetiva, Gheorghe Roșca Codreanu National College, Bârlad, Romania.

507. Calculate the integral

$$\int_1^\infty \frac{\ln x}{x^3 + x\sqrt{x} + 1} dx.$$

Proposed by Vasile Mircea Popa, Lucian Blaga University, Sibiu, Romania.

508. Let K be a field and let $A, B \in M_n(K)$ with $n \geq 1$ such that $AB - BA = c(A - B)$ for some $c \in K \setminus \{0\}$. (i) Prove that if $\text{char } K = 0$ or $\text{char } K > n$, then A and B have the same eigenvalues. (ii) Prove that if $0 < \text{char } K \leq n$ then (i) is no longer true.

Proposed by Constantin-Nicolae Beli, IMAR, București, Romania.

509. Let m and n be positive integers and let $A_1, \dots, A_m \in \mathcal{M}_n(\mathbb{R})$. For every $i \in \{1, \dots, m\}$ denote by $\lambda_{i1}, \dots, \lambda_{in} \in \mathbb{C}$ the eigenvalues of A_i . Prove that there exist $\varepsilon_1, \dots, \varepsilon_m \in \{-1, 1\}$ such that the eigenvalues $\mu_1, \dots, \mu_n \in \mathbb{C}$ of the matrix $\varepsilon_1 A_1 + \dots + \varepsilon_m A_m \in \mathcal{M}_n(\mathbb{R})$ satisfy

$$\sum_{j=1}^n \mu_j^2 \geq \sum_{i=1}^m \sum_{j=1}^n \lambda_{ij}^2.$$

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Romania.

510. Prove that

$$\sum_{n=1}^{\infty} \binom{4n}{2n} \frac{1}{16^n n^2 (2n+1)} = 4 \text{Li}_2\left(\frac{1-\sqrt{2}}{2}\right) + \frac{\pi^2}{3} + 4 \log\left(\frac{1+\sqrt{2}}{4}\right) - 2 \log^2\left(\frac{1+\sqrt{2}}{2}\right) - \log^2(4) + 4(\sqrt{2}-1),$$

where $\text{Li}_2(x) = -\int_0^x \frac{\log(1-t)}{t} dt$ is the dilogarithm.

Proposed by Seán M. Stewart, Bomaderry, NSW, Australia.

511. Find the best lower and upper bounds for $\sum_{i=1}^n \cos(\angle A_i)$ over all convex n -gons $A_1 A_2 \dots A_n$.

Proposed by Leonard Giugiu, Traian National College, Drobeta Turnu Severin, Romania and Florin Visescu, Mihai Eminescu National College, București, Romania.

512. Evaluate the series

$$\sum_{n=1}^{\infty} \left(\left(n \left(n \sum_{k=n}^{\infty} \frac{1}{k^2} - 1 \right) - \frac{1}{2} \right) - \frac{1}{6} \right).$$

Proposed by Marian Tetiva, Gheorghe Roșca Codreanu National College, Bârlad, Romania.

513. Find all differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which verify the identity

$$xf'(x) + kf(-x) = x^2 \quad \text{for all } x \in \mathbb{R},$$

where $k \geq 1$ is an integer.

Proposed by Vasile Pop and Ovidiu Furdul, Technical University of Cluj-Napoca, Romania.

514. Evaluate the integral

$$\int_0^1 \frac{\log^n x}{\sqrt{x(1-x)}} dx,$$

where n is a positive integer.

Proposed by Mircea Ivan, Technical University of Cluj-Napoca, Romania.

515. Let $S = \{(\alpha, \beta, \gamma) \in (0, \pi/2)^3 : \alpha + \beta + \gamma = \pi\}$. On S define the real valued functions

$$a := a(\alpha, \beta, \gamma) = \sqrt{\sin^4 \beta + \sin^4 \gamma - 2 \sin^2 \beta \sin^2 \gamma \cos 2\alpha},$$

$$b := b(\alpha, \beta, \gamma) = \sqrt{\sin^4 \alpha + \sin^4 \gamma - 2 \sin^2 \alpha \sin^2 \gamma \cos 2\beta},$$

$$c := c(\alpha, \beta, \gamma) = \sqrt{\sin^4 \alpha + \sin^4 \beta - 2 \sin^2 \alpha \sin^2 \beta \cos 2\gamma}.$$

Prove that the function $f : S \rightarrow S$ given by

$$f(\alpha, \beta, \gamma) = \left(\arccos \frac{b^2 + c^2 - a^2}{2bc}, \arccos \frac{a^2 + c^2 - b^2}{2ac}, \arccos \frac{a^2 + b^2 - c^2}{2ab} \right)$$

is well defined. Is f injective? Is it surjective?

Proposed by Leonard Giugiu, Colegiul Național Traian, Drobeta Turnu Severin, Romania, and Abdilkadir Altıntaş, Emirdağ, Afyonkarahisar, Turkey.

516. Let $n \geq 2$ be an integer. Let $v_1, \dots, v_{n-1}$ be some orthonormal vectors and let v be a unit vector in $\mathbb{R}^n$. We regard $v_1, \dots, v_{n-1}, v$ as column vectors. Consider the $n \times n$ matrix

$$A = v_1 v_1^T + \dots + v_{n-1} v_{n-1}^T - v v^T.$$

If A is not invertible, prove that $A^2 = A$ and determine its rank.

Proposed by Marian Pantiruc, Gheorghe Asachi Technical University of Iași, Romania.

517. Calculate the sum

$$S = \sum_{p,q,r=1}^{\infty} \frac{3p+r}{5^{p+q+r} r (p+q)(q+r)(r+p)}.$$

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Romania.

518. Calculate the integral:

$$\int_0^{\infty} \frac{x^2 \sqrt{x} \ln x}{x^4 + x^2 + 1} dx.$$

Proposed by Vasile Mircea Popa, Lucian Blaga University, Sibiu, Romania.

519. Calculate

$$\sum_{n=1}^{\infty} (-1)^n n^2 \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} - \ln 2 + \frac{1}{4n} - \frac{1}{16n^2} \right).$$

Proposed by Ovidiu Furdul and Alina Sîntămărian, Technical University of Cluj-Napoca, Romania.

520. Let $(x_n)_{n \geq 0}$ be a sequence with $x_0 \in (0, \pi/2)$ and

$$x_{n+1} = \begin{cases} \sin(\cos x_n) & \text{if } n \text{ is even,} \\ \cos(\sin x_n) & \text{if } n \text{ is odd.} \end{cases}$$

Prove that $x_{2n} \rightarrow a$ and $x_{2n+1} \rightarrow b$ as $n \rightarrow \infty$, where a, b are constants independent of the choice of x_0 . Also determine if the series $\sum_{n=1}^{\infty} |x_{2n} - a|^\alpha$ and $\sum_{n=1}^{\infty} |x_{2n+1} - b|^\alpha$ are convergent for $\alpha > 0$.

Proposed by Radu Strugariu, Gheorghe Asachi Technical University of Iași, Romania.

521. Prove that

$$\lim_{p \rightarrow \infty} \sum_{i,j=0}^p \frac{(-4a^2)^{i+j}}{2i+2j+1} \binom{p+i}{p-i} \binom{p+j}{p-j} = \frac{1}{4a} \ln \left(\frac{1+a}{1-a} \right) \quad \forall a \in (0,1).$$

Proposed by Florin Stănescu, Șerban Cioculescu School, Găești, Dâmbovița, Romania.

522. Evaluate the integral

$$\int_0^\infty \frac{\log(1+x^{10})}{1+x^2} dx.$$

Proposed by Seán M. Stewart, King Abdullah University of Science and Technology (KAUST), Saudi Arabia.

523. Prove that

$$\text{rank}(A - ABA) - \text{rank}(B - BAB) = \text{rank}(A) - \text{rank}(B)$$

for all $A, B \in \mathcal{M}_n(\mathbb{C})$.

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Romania, and Mihai Opincariu, Avram Iancu National College, Brad, Romania.

524. Determine all bijective and differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$, with $f'(x) \neq 0$ for all $x \in \mathbb{R}$, satisfying

$$f(x) + f\left(\frac{1}{f'(x)} - x\right) = 1 \quad \text{for every } x \in \mathbb{R}.$$

Proposed by Mircea Rus, Technical University of Cluj-Napoca, Romania.

525. Let $n \geq 4$ and let $a_1, \dots, a_n > 0$. (i) Prove that if $\frac{1}{a_1} + \dots + \frac{1}{a_n} \leq 1$, then

$$2 \sum_{1 \leq i < j \leq n} a_i a_j - n^3(n-1) \geq 2(n-1)^2(a_1 + \dots + a_n - n^2).$$

(ii) Prove that if $\frac{1}{a_1} + \dots + \frac{1}{a_n} \geq 1$ and $k = \frac{n^2-n+1}{(n-1)^3}$, then

$$a_1^2 + \dots + a_n^2 - n^3 \geq k \left(2 \sum_{1 \leq i < j \leq n} a_i a_j - n^3(n-1) \right).$$

Proposed by Vasile Cîrtoaje, Petroleum-Gas University of Ploiești, Romania, and Leonard Giugiu, Traian National College, Drobeta-Turnu Severin, Romania.

526. If $a, b \in \mathbb{C}$ and $k \geq 2$, define

$$\mathcal{F}_{a,b}^k = \{f : \mathbb{C} \rightarrow \mathbb{C} \mid f^{(k)}(z) = az + b\},$$

where $f^{(k)} = f \circ \dots \circ f$ (k times). For given $a, b \in \mathbb{C}$ and $k \geq 2$ find necessary and sufficient conditions such that a set $A \subseteq \mathbb{C}$ can be written as $A = \text{Fix}(f)$ for some $f \in \mathcal{F}_{a,b}^k$.

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Romania, and Constantin-Nicolae Beli, IMAR, Bucharest, Romania.

527. For every non-negative integer n , evaluate the integral

$$I_n = \int_0^\infty \frac{\log^n(x) \sin(x)}{x} dx.$$

The answer may be given either in a closed form or recursively. It may also include values of the zeta function at integers ≥ 2 .

Proposed by Seán M. Stewart, King Abdullah University of Science and Technology (KAUST), Saudi Arabia.

528. If $a_1, a_2, a_3, a_4, a_5, a_6$ are non-negative real numbers such that

$$a_1 a_2 + a_2 a_3 + a_3 a_4 + a_4 a_5 + a_5 a_6 + a_6 a_1 = 6, \quad a_1 \geq a_2 \geq a_3 \geq a_4 \geq a_5 \geq a_6,$$

then

$$\frac{1}{a_1+3} + \frac{1}{a_2+3} + \frac{1}{a_3+3} + \frac{1}{a_4+3} + \frac{1}{a_5+3} + \frac{1}{a_6+3} \geq \frac{3}{2}.$$

Proposed by Vasile Cîrtoaje, Petroleum-Gas University of Ploiești, Romania.

529. Prove that

$$\prod_{n=2}^{\infty} e^{n^2+1/2} \left(1 - \frac{1}{n^2}\right)^{n^4} = \pi \exp\left(-\frac{7}{4} + \frac{3\zeta(3)}{\pi^2}\right).$$

Proposed by Moubinool Omarjee, Lycée Henri IV, Paris, France.

530. Let $a, b, c, d, e \in \mathbb{R}$ such that $a \geq b \geq c \geq d \geq e \geq 0$ and

$$ab + bc + cd + de + ea = 5.$$

Prove that

$$(3-a)^2 + (3-b)^2 + (3-c)^2 + (3-d)^2 + (3-e)^2 \geq 20.$$

Proposed by Vasile Cîrtoaje, Petroleum-Gas University of Ploiești, Romania.

531. If $n \geq 2$ is an integer, it is known that if $k \nmid n$ for all $k \leq \sqrt{n}$, then n is prime. Let $\theta \in [1/3, 1/2]$. Determine the probability for a number $n \geq 2$ with $k \nmid n$ for all $k \leq n^\theta$ to be a prime. That is, determine the limit

$$\lim_{x \rightarrow \infty} \frac{\#\{p \leq x : p \text{ prime}\}}{\#\{n \leq x : k \nmid n \ \forall k \leq n^\theta\}}.$$

Proposed by Constantin-Nicolae Beli, IMAR, București, Romania.

532. Let $(S, \cdot)$ be a semigroup with the property that for every $x \in S$ there is a unique $x' \in S$ such that $(xx')^2 = xx'$. Prove that S is a group.

Proposed by Gheorghe Andrei, Constanța, and Mihai Opincariu, Brad, Romania.

533. Prove that

$$\sum_{m=2}^{\infty} (-1)^m (\zeta(m) - \zeta(m+1)) \left(H_{\frac{m-1}{2}} - H_{\frac{m}{2}}\right) = \frac{\pi^2}{3} (1 - \log 2) - 2\gamma,$$

where $\zeta(k) = \sum_{n=1}^{\infty} 1/n^k$, $H_n = \int_0^1 \frac{1-x^n}{1-x} dx$ and γ is the Euler-Mascheroni constant.

Proposed by Narendra Bhandari, Bajura, and Yogesh Joshi, Kailali, Nepal.

534. Let $n \geq 4$. (a) Find the smallest positive constant k_n for which the inequality

$$\frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}} + k_n (\sqrt{a_1} - \sqrt{a_n})^2 \geq \frac{a_1 + \dots + a_n}{n}$$

holds for all $a_1, \dots, a_n \in \mathbb{R}$ with $a_1 \geq \dots \geq a_n > 0$. (b) Find the largest positive constant c_n for which the inequality

$$\frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}} + c_n (\sqrt{a_1} - \sqrt{a_n})^2 \leq \frac{a_1 + \dots + a_n}{n}$$

holds for all $a_1, \dots, a_n \in \mathbb{R}$ with $a_1 \geq \dots \geq a_n > 0$.

Proposed by Leonard Giugiu, Traian National College, Drobeta-Turnu Severin, and Vasile Cîrtoaje, Petroleum-Gas University of Ploiești, Romania.

535. Let $A, B \in M_n(\mathbb{C})$ be two matrices of the same rank and let $k \in \mathbb{N}^*$ (here $\mathbb{N}^* = \mathbb{Z}_{\geq 1}$). Then

$$A^{k+1}B^k = A \quad \text{if and only if} \quad B^{k+1}A^k = B.$$

Proposed by Vasile Pop, Cluj-Napoca, and Mihai Opincariu, Brad, Romania.

536. Let p be a prime number, $\mathbb{F}_p$ the field with p elements, and $n \geq 1$ an integer. If $f \in \mathbb{F}_p[X]$ is the polynomial $X^p - X \in \mathbb{F}_p[X]$ composed with itself n times, determine the splitting field of f over $\mathbb{F}_p$.

Proposed by Tudor Pășanu, École Polytechnique, Paris, France.

537. Let $A, B \in \mathcal{M}_n(\mathbb{C})$ be such that $A^2 = A$ and $B^2 = B$. Prove that

$$\text{Im}(AB - BA) = \text{Im}(A + B - I_n) \cap \text{Im}(A - B),$$

where $\text{Im } M = \{MX \mid X \in \mathcal{M}_{n,1}(\mathbb{C})\}$ for every $M \in \mathcal{M}_n(\mathbb{C})$.

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Romania.

538. Let $n \geq 1$ be an integer and let $a_1, \dots, a_{2n} \in \mathbb{Z}$ be pairwise distinct. Prove that

$$\sum_{i=1}^{2n} a_i^2 + \left(\sum_{i=1}^{2n} a_i\right)^2 \geq \frac{n(n+1)(2n+1)}{3}.$$

When does equality hold?

Proposed by Leonard Giugiu, Traian National College, Drobeta-Turnu Severin, Romania.

539. Let $n \geq 1$ be an integer and let $X = \{1, \dots, n\}$. Denote by F_X the set of all functions $f : X \rightarrow X$ and by S_X the symmetric group on X . If $f, g \in F_X$, we say that f and g are conjugate and write $f \sim g$ if there is $\sigma \in S_X$ such that $g = \sigma f \sigma^{-1}$. Let M_X be the set of all $f \in F_X$ such that for every $\emptyset \neq Y \subseteq X$ with $f(Y) \subseteq Y$ we have $f(Y) = f(X)$. (i) Prove that if $f \in M_X$ and $g \sim f$ then $g \in M_X$. (ii) Prove that $|M_X/\sim| = \frac{1}{n} \sum_{d|n} \phi(d) 2^{n/d} - 1$. Proposed by Constantin-Nicolae Beli, IMAR, Bucharest, Romania.

540. For any matrix M , denote $M^* = \overline{M}^t$ the conjugate transpose of M . Let $A, B \in \mathcal{M}_n(\mathbb{C})$ be such that $A^*B = O_n$. Prove that $\text{rank}(A^*A + B^*B) \leq \text{rank}(AA^* + BB^*)$.

Proposed by Mihai Opincariu, Brad, Romania, and Vasile Pop, Technical University of Cluj-Napoca, Romania.

541. Calculate

$$\sum_{n=1}^{\infty} \frac{H_n H_{n+1}}{(2n+1)(2n+3)},$$

where $H_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}$ denotes the n -th harmonic number.

Proposed by Ovidiu Furdui and Alina Sintămărian, Technical University of Cluj-Napoca, Romania.

542. Let $n \geq 2$ be an integer and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be given by $f(x_1, \dots, x_n) = 0$ if $(x_1, \dots, x_n) = (0, \dots, 0)$ and otherwise

$$f(x_1, \dots, x_n) = \frac{\sqrt[5]{x_1^4 \cdots x_n^4}}{\sqrt[3]{x_1^2 \cdots x_n^2} + (x_2 - x_1)^2 + (x_3 - x_1)^2 + \cdots + (x_n - x_1)^2}.$$

Prove that: (i) f is continuous at $(0, \dots, 0)$. (ii) f is Fréchet differentiable at $(0, \dots, 0)$ if and only if $n \geq 8$.

Proposed by Dumitru Popa, Department of Mathematics, Ovidius University of Constanța, Romania.

544. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$ such that $f(0) = f(1/2) = f(1) = 0$. Show that

$$\int_0^1 (f'(x))^2 dx \geq 48 \left(\int_0^1 f(x) dx \right)^2.$$

Proposed by Robert Dragomirescu, Stanford University, USA, and Cezar Lupu, Yanqi Lake Beijing Institute of Mathematical Sciences and Applications (BIMSA) and Tsinghua University, P. R. China.

545. Let $A, B \in M_n(\mathbb{R})$ such that $A^2 = -I_n$, $\det B \neq 0$, and $AB = -BA$. Prove that n is even and the sign of $\det B$ is $(-1)^{n/2}$.

Proposed by Mihai Opincariu, Brad, and Vasile Pop, Cluj-Napoca, Romania.

546. Let $X, Y \in M_n(\mathbb{C})$ such that $Y^2 = YX - XY$ and the rank of $X + Y$ is 1. Prove that $Y^3 = YXY = O_n$.

Proposed by Stănescu Florin, Șerban Cioculescu School, Găești, Romania.

547. Prove that

$$\int_0^{\infty} \frac{|\sin x|}{1+x^2} dx = \frac{e^2 - 1}{2e} \ln \left(\frac{e+1}{e-1} \right).$$

Proposed by Vasile Mircea Popa, Lucian Blaga University, Sibiu, Romania.

548. Let $A \in M_n(\mathbb{C})$ such that $(I_n - AA^*)^2 = I_n - A^*A$. Prove that $A^2A^* = A$. (Here $A^* = \bar{A}^t$ denotes the conjugate transpose of A .)

Proposed by Mihai Opincariu, Brad, and Vasile Pop, Cluj-Napoca, Romania.

549. Let $n \geq 3$ and let $a_1, \dots, a_n \in \mathbb{Z}_{\geq 0}$ be pairwise distinct. Denote by s_1, s_2, s_3 the first symmetric sums in the variables $a_1, \dots, a_n$, i.e. $s_1 = \sum_i a_i$, $s_2 = \sum_{i < j} a_i a_j$, and $s_3 = \sum_{i < j < k} a_i a_j a_k$. Prove that

$$(n-2)s_1 \left(s_2 - \frac{n(n-1)(n+1)}{12} \right) \geq 3ns_3.$$

When does equality hold?

Proposed by Leonard Giugiu, Traian National College, Drobeta-Turnu Severin, Romania.

550. Let $a_1, a_2, \dots, a_n$ be real numbers such that $a_1 \geq a_2 \geq \dots \geq a_n \geq 0$ and $a_1 a_2 + a_2 a_3 + \dots + a_n a_1 = n$, and let $E_n(k) = a_1^k + a_2^k + \dots + a_n^k$. (a) Prove that $E_5(k) \geq 5$ for $k \geq \frac{5}{4}$. (b)* Prove or disprove that $E_7(k) \geq 7$ for $k \geq \frac{3}{2}$. (This is an open problem. At this time, the author doesn't have a solution.)

Proposed by Vasile Cîrtoaje, Petroleum-Gas University of Ploiești, Romania.

551. Solve in $\mathcal{M}_2(\mathbb{R})$ the equation $A^{2024} = -A^T$, where A^T denotes the transpose of A .

Proposed by Ovidiu Furdui and Alina Sintămărian, Technical University of Cluj-Napoca, Romania.

552. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function which is derivable on $(0, 1]$, such that $f'(x) < 0$ for $x \in (0, 1]$, $f(1) = 0$ and $f'(1) < 0$. Prove that for every integer $n \geq 1$ the equation $f(x) = x^n$ has a unique solution in the interval $(0, 1)$, denoted by a_n , and that $\lim_{n \rightarrow \infty} \frac{n}{\ln n}(a_n - 1) = -1$.

Proposed by Dumitru Popa, Department of Mathematics, Ovidius University of Constanța, Romania.

553. Let $ABCD$ be an isosceles tetrahedron with centroid G . Let M, N be two points such that $\overrightarrow{NG} = 3\overrightarrow{GM}$. Prove that

$$NA + NB + NC + ND \geq MA + MB + MC + MD.$$

Proposed by Leonard Giugiu, Drobeta Turnu-Severin, Romania.

554. Let $n \in \mathbb{N}$, $n \geq 2$. (a) Prove that $\det(A^2 - B^2)(C^2 - B^2) \geq 0$ for all $A, B, C \in \mathcal{M}_n(\mathbb{R})$ with $AB = BC$. (b) Find all values $k \geq 1$ such that $\det(A^k - B^2)(C^k - B^2) \geq 0$ holds for all $A, B, C \in \mathcal{M}_n(\mathbb{R})$ with $AB = BC$.

Proposed by Mihai Opincariu, Brad, Romania, and Vasile Pop, Technical University of Cluj-Napoca, Romania.

555. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative such that $f(1) = 0$ and $f'(1) = 1$. Prove that there exists $c \in (0, 1)$ such that

$$f(c) = f'(c) \int_0^c f(x) dx.$$

Proposed by Cezar Lupu, Beijing Institute of Mathematical Sciences and Applications (BIMSA) and Tsinghua University, Beijing, P. R. China.

556. For given $n \geq 3$, prove that $k = 2n - 3$ is the smallest positive constant such that

$$\frac{1}{a_1 + k} + \frac{1}{a_2 + k} + \cdots + \frac{1}{a_n + k} \leq \frac{n}{1 + k}$$

holds for any nonnegative real numbers $a_1, \dots, a_n$ such that at most one of them is > 1 and $\sum_{1 \leq i < j \leq n} a_i a_j = \frac{n(n-1)}{2}$.

Proposed by Vasile Cîrtoaje, Petroleum-Gas University of Ploiești, Romania.

557. Find the differentiable functions $f : (0, \infty) \rightarrow \mathbb{R}$ that satisfy the identity:

$$f'(x) = x \cdot f\left(\frac{1}{x}\right) \quad \text{for all } x \in (0, \infty).$$

Proposed by Dorian Popa, Technical University of Cluj-Napoca, Romania.

558. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^1 x^k f(x) dx = 0 \quad \text{for } 0 \leq k \leq n-1, \quad \int_0^1 x^n f(x) dx = 1.$$

Prove that

$$\int_0^1 f^2(x) dx \geq (2n+1) \binom{2n}{n}^2.$$

Proposed by Cezar Lupu, Beijing Institute of Mathematical Sciences and Applications (BIMSA) and Tsinghua University, Beijing, P. R. China.

559. Let $f : [0, 1] \rightarrow [-1, 1]$ be a continuous function, with finite derivative in 0 and $f(0) = 1$. Find $\lim_{n \rightarrow \infty} \int_0^1 f^n(x^n) dx$.

Proposed by Mircea Rus, Technical University of Cluj-Napoca, Romania.

560. Let $(x_n)_{n \geq 1}$ be the sequence defined by $x_1 \in (0, 1)$ and $x_{n+1} = x_n - \frac{x_n^2}{2n}$ for all $n \geq 1$. Prove that the sequence $(x_n)_{n \geq 1}$ is convergent to a limit $C > 0$ and moreover,

$$\lim_{n \rightarrow \infty} 8^{n-1} \left(x_n - C - \frac{C^2}{2^{n-1}} - \frac{C^3}{3 \cdot 4^{n-2}} \right) = \frac{12C^4 + 32C^3}{21}.$$

Proposed by Dumitru Popa, Ovidius University of Constanța, Romania.

561. Calculate

$$\sum_{n=1}^{\infty} \left[n^2 \left(\frac{1}{n^3} - \frac{1}{(n+1)^3} + \frac{1}{(n+2)^3} - \cdots \right) - \frac{1}{2n} \right].$$

Proposed by Ovidiu Furdul and Alina Sintămărian, Technical University of Cluj-Napoca, Romania.

562. For any matrix M , let $M^* = \overline{M}^t$ denote the conjugate transpose of M . The matrix M is called anti-Hermitian if $M^* = -M$. Prove that if $A \in \mathcal{M}_n(\mathbb{C})$ is invertible and anti-Hermitian, then the function

$$f : \mathcal{M}_n(\mathbb{C}) \rightarrow \mathcal{M}_n(\mathbb{C}), \quad f(X) = AX - XA^2, \quad X \in \mathcal{M}_n(\mathbb{C}),$$

is bijective.

Proposed by Mihai Opincariu, Brad, Romania, and Vasile Pop, Technical University of Cluj-Napoca, Romania.

563. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative such that $f(0) = f(1)$ and $\int_0^1 xf(x)dx = 0$. Prove that

$$\int_0^1 (f'(x))^2 dx \geq 180 \left(\int_0^1 f(x) dx \right)^2.$$

Proposed by Cezar Lupu, Beijing Institute of Mathematical Sciences and Applications (BIMSA), Tsinghua University, Beijing, P. R. China, and Tudorel Lupu, Decebal High School, Constanța, Romania.

564. Let $a_1, \dots, a_7$ be nonnegative real numbers such that $a_1a_2 + a_2a_3 + \dots + a_7a_1 = 7$. Prove that

$$(a) \quad \frac{1}{5a_1 + 3} + \frac{1}{5a_2 + 3} + \dots + \frac{1}{5a_7 + 3} \geq \frac{7}{8},$$

$$(b)^* \quad \frac{1}{a_1 + 1} + \frac{1}{a_2 + 1} + \dots + \frac{1}{a_7 + 1} \geq \frac{7}{2}.$$

Proposed by Vasile Cîrtoaje, Petroleum-Gas University of Ploiești, Romania.

565. Let $f \in C^{2n+1}([0, 1], \mathbb{R})$ for some $n \geq 0$ such that $f(1/2) = f'(1/2) = f''(1/2) = \dots = f^{(2n)}(1/2) = 0$. Prove that

$$\int_0^1 (f^{(2n+1)}(x))^2 dx - \left(\int_0^1 f^{(2n+1)}(x) dx \right)^2 \geq 2^{4n+2} (4n+3) ((2n+1)!)^2 \left(\int_0^1 f(x) dx \right)^2.$$

Proposed by Florin Stănescu, Șerban Cioculescu School, Găești, Romania.

566. Prove that for all natural numbers $n \geq 4$ the equation

$$\sum_{k=1}^n \frac{1}{k^2 + 2k + x} = \frac{3}{4}$$

has a unique solution in the interval $(-1, 0)$, denoted by x_n , and find the value of $\lim_{n \rightarrow \infty} nx_n$.

Proposed by Dumitru Popa, Department of Mathematics, Ovidius University of Constanța, Romania.

567. Let $n \geq 4$ and let $a_1, \dots, a_n \geq 0$ such that

$$a_1^2 + \dots + a_n^2 + (n^2 - 3n + 1)a_1 \dots a_n \geq (n-1)^2.$$

Prove that

$$a_1 + \dots + a_n + \frac{1}{n-1} \sqrt{\frac{1}{n-1} \sum_{1 \leq i \leq j \leq n} (a_i - a_j)^2} \geq n.$$

Proposed by Leonard Giugiu, Greci School, Mehedintți, Romania.

568. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a twice differentiable function, with f'' continuous, such that $f(0) = f(1)$. Prove that there exists $\xi \in (0, 1)$ such that

$$\xi^2 f(\xi) = 2 \int_0^\xi xf(x)dx.$$

Proposed by Cezar Lupu, Beijing Institute of Mathematical Sciences and Applications (BIMSA) and Tsinghua University, Beijing, P.R. China.

569. For an odd positive integer x put $x!! = x(x-2) \dots 3 \cdot 1$. 1) Let a and b be positive real numbers. Calculate

$$\lim_{n \rightarrow \infty} \left(\sqrt[n]{a \cdot (2n+1)!!} - \sqrt[n-1]{b \cdot (2n-1)!!} \right).$$

2) Let $f, g \in \mathbb{R}[x]$ be polynomials of the same degree with the coefficient of the leading term positive. Calculate

$$\lim_{n \rightarrow \infty} \left(\sqrt[n]{(2n+1)!!} f(n) - \sqrt[n-1]{(2n-1)!!} g(n-1) \right).$$

Proposed by Ovidiu Furdui and Alina Sintămărian, Technical University of Cluj-Napoca, Romania.

570. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative such that $f(0) = f(1) = 0$ and $\int_0^1 xf(x) dx = 0$. Prove that

$$\int_0^1 (f'(x))^2 dx \geq 192 \left(\int_0^1 f(x) dx \right)^2.$$

Proposed by Ulrich Abel and Vitaliy Kushnirevych, Technische Hochschule Mittelhessen, Germany.

571. Let $a_1, a_2, a_3, a_4 \geq 0$ be real numbers. Prove that

$$a_1 + a_2 + 3a_3 + 3a_4 + \sqrt{(a_1 - a_3)^2 + (a_2 - a_4)^2} \geq \sum_{k=1}^4 \sqrt{a_k^2 + a_{k+1}^2},$$

where $a_5 = a_1$.

Proposed by Leonard Giugiac, Middle School Greci, Mehedinți, Romania.

572. Prove that there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ the equation

$$n^2 \iint_{[0,1]^2} x^n y^n e^{xy^2} dx dy = e$$

has a unique real solution denoted by z_n and, moreover, $\lim_{n \rightarrow \infty} n(z_n - 1) = 4$.

Proposed by Dumitru Popa, Department of Mathematics, Ovidius University of Constanța, Romania.

573. Let $B = \bigcup_{n \geq 1} B_n$, where

$$B_n = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_i \leq x_{i+2} \text{ for all } 1 \leq i \leq n-2\}.$$

On B we define the relations $\leq$ and $<$ as follows. Let $x, y \in B$ with $x = (x_1, \dots, x_m)$ and $y = (y_1, \dots, y_n)$. It is known that $(B, \leq)$ is partially ordered. We say that $x < y$ if $x \leq y$, $m - n \leq 2$, and for every $1 < i < m$ we have either $x_{i+1} \geq y_{i-1}$ or $i \leq n$ and $x_i + x_{i+1} = y_{i-1} + y_i$. If $x < y$ we denote

$$\delta(x, y) = \sum_{i=1}^n (-1)^i (x_i - y_i)_+,$$

where $x_+ = \frac{x+|x|}{2}$ is the positive part of x . Prove that if $x, y, z \in B$ satisfy $x \leq y \leq z$ and $x < z$, then $x < y < z$ and $\delta(x, y) + \delta(y, z) = \delta(x, z)$.

Proposed by Constantin-Nicolae Beli, IMAR, București, Romania.

574. The matrices $A, B \in \mathcal{M}_n(\mathbb{C})$, $n \geq 2$, satisfy the relations:

$$(1) A^2 = B^2; \quad (2) A^3 + BAB = 2I_n.$$

(a) Prove that $A^3 = I_n$.

(b) Prove that if A and B have no common eigenvalues, then $A + B = O_n$.

Proposed by Vasile Pop and Constantin-Cosmin Todea, Technical University of Cluj-Napoca, Romania.

575. Let n be a positive integer with $n \geq 3$. Prove that $\frac{n}{4}$ is the least positive value of the constant k such that the inequality

$$a_1 + a_2 + \dots + a_n + \frac{k(a_1 - a_n)^2}{a_1 + a_n} \geq \sqrt{n(a_1^2 + a_2^2 + \dots + a_n^2)}$$

holds whenever $a_1 \geq a_2 \geq \dots \geq a_n \geq 0$.

Proposed by Vasile Cîrtoaje, Petroleum-Gas University of Ploiești, Romania, and Leonard Giugiu, Middle School Greci, Mehedinți, Romania.

576. We say that $x \in \mathbb{R}$ is a *nice number* if $\sum_{k=1}^{\infty} k^n x^k$ is an integer for all $n \in \mathbb{N}$, $n \geq 1$.

(a) Find all the nice numbers.

(b) If x is a nice number, show that $\sum_{k=1}^{\infty} k^n x^k$ is an even natural number for all $n \in \mathbb{N}$, $n \geq 1$.

Proposed by Mircea Rus, Technical University of Cluj-Napoca, Romania.

577. Let $a \in \mathbb{R} \setminus \{0\}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ a differentiable function with the property

$$|f'(x) + af(x)| \leq 1 \quad \text{for all } x \in \mathbb{R}.$$

Prove that there exists a unique differentiable function $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$g'(x) + ag(x) = 0 \quad \text{and} \quad |f(x) - g(x)| \leq \frac{1}{|a|} \quad \text{for all } x \in \mathbb{R}.$$

Proposed by Dorian Popa, Technical University of Cluj-Napoca, Romania.

578. Let $A \in \mathcal{M}_{n,m}(\mathbb{C})$, $B \in \mathcal{M}_{m,n}(\mathbb{C})$ and $C \in \mathcal{M}_m(\mathbb{C})$ such that $\text{rank } A = \text{rank } B = m$. Show that there exists a unique matrix $D \in \mathcal{M}_n(\mathbb{C})$ such that $BDA = C$ and determine an explicit formula for D .

Proposed by Vasile Pop, Technical University of Cluj-Napoca, Romania.

579. Let $a \in (0, 1)$. Find all continuously differentiable functions $f : [0, \infty) \rightarrow \mathbb{R}$ with $f(0) = 0$ such that

$$(x - y)(f'(x - y))^2 \leq f(x - y) \cdot f'(ax) \quad \text{for all } x, y \text{ with } x > y \geq 0.$$

Proposed by Stanescu Florin, Serban Cioculescu Secondary School, Gaesti, Dambovita, Romania.

580. Prove that $3/4$ is the largest positive value of the power exponent k such that the inequality

$$(ab)^k + (ac)^k + (ad)^k + (bc)^k + (bd)^k + (cd)^k \leq 6$$

holds for any $a \geq b \geq c \geq d \geq 0$ with $ab + bc + cd + da = 4$.

Proposed by Vasile Cirttoaje, Petroleum-Gas University of Ploiesti, Romania.

581. Let $(x_n)_{n \geq 1}$ be the sequence defined by $x_1 \in (0, 1)$ and

$$x_{n+1} = \frac{x_1(1 - x_1^2) + x_2\sqrt{1 - x_2^2} + \cdots + x_n\sqrt{1 - x_n^2}}{n} \quad \text{for all } n \geq 1.$$

Prove that the sequence $(x_n)_{n \geq 1}$ is convergent to a limit $C > 0$ and, moreover,

$$\lim_{n \rightarrow \infty} n(x_n - C) = -C \ln(1 - C^2).$$

Proposed by Dumitru Popa, Department of Mathematics, Ovidius University of Constanta, Romania.

582.

(a) Let $(V, \langle \cdot, \cdot \rangle)$ be an Euclidean space over $\mathbb{C}$ and $T : V \rightarrow V$ an endomorphism with the property that $\langle T(z), z \rangle = 0$ for all $z \in V$. Prove that $T = 0$.

Proposed by ...

583. Let $n \geq 3$ be an integer. Find the largest real constant C with the property that

$$\frac{a_1 + a_2 + \cdots + a_n}{\sqrt[n]{a_1 a_2 \cdots a_n}} - n \geq C \cdot \frac{(\sqrt[n]{a_1} - \sqrt[n]{a_n})^2}{\sqrt[n]{a_1 a_n}}$$

holds for all positive real numbers $a_1 \geq a_2 \geq \cdots \geq a_n$.

Proposed by Leonard Giugiu, Greci Secondary School, Mehedinti, Romania.

584. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function with $\int_0^1 f(x) dx = 0$ and let $\alpha \in \mathbb{R}$, $\alpha > 0$. Prove that there is $c \in (0, 1)$ such that

$$\int_0^c x^{\alpha+2} f(x) dx = -\alpha c^{\alpha+1} \int_0^c x f(x) dx.$$

Proposed by Stanescu Florin, Serban Cioculescu Secondary School, Gaesti, Dambovita, Romania.